



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

I/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1101	English – I *	3	3	1	--	4	30	70	100
B17 BS 1102	Mathematics – I *	3	3	1	--	4	30	70	100
B17 BS 1105	Engineering Chemistry	3	3	1	--	4	30	70	100
B17 ME 1101	Engineering Mechanics	3	3	1	--	4	30	70	100
B17 ME 1102	Engineering Drawing	3	1	--	3	4	30	70	100
B17 CE 1101	Environmental Studies *	2	2	1	--	3	30	70	100
B17 BS 1107	Engineering Chemistry Lab	2	--	--	3	3	50	50	100
B17 BS 1108	English Communication Skills Lab – I *	2	--	--	3	3	50	50	100
B17 BS 1109	Engineering Workshop & IT Workshop	2	--	--	3	3	50	50	100
B17 BS 1111	Inner Engineering	--	--	--	2	2	--	--	--
Total		23	15	5	14	34	330	570	900

* Common to both Group - A and Group - B

Code: B17 BS 1101

ENGLISH - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop the communication skills of the students in both the formal and informal situations.
4. To expose the students to various forms of writing in formal settings in order to build confidence in standard grammar
5. To introduce the students various types of reading comprehension practices.

Course Outcomes: At the end of the Course, Student will be able to:

1. Understand the rudiments of LSRW Skills, comprehension and fluency of speech.
2. Gain confidence and competency in vocabulary and grammar.
3. Listen, speak, read and write effectively in both the academic and non- academic environment.
4. Extend his/her reading skills towards literature.
5. Strengthen his/her analytical and compositional skills.

SYLLABUS

Life through Language: An Effective Learning Experience

Life through Language has a systematic structure that builds up communicative ability progressively through the chapters. It will enable the learner to manage confusion; frame question for themselves and others; develop new ideas; support ideas with evidence; express themselves with poise and clarity; and think critically. Acquisition of skill leads to confidence.

UNIT-I

People and Places:- Word search - Ask yourself-Self-assessment-I -Self-assessment-II - Sentence and its types- Describing people, places and events-Writing sentences-Self-awareness-Self-motivation, Dialogue writing.

UNIT-II

Personality and Lifestyle:- Word quiz – Verbs-Adverbs-Negotiations-Proving yourself-Meeting Carl Jung- Describing yourself- Living in the 21st century- Using your dictionary-Communication-Adaptability.

UNIT-III

Media and Environment: - A list of 100 basic words – Nouns- Pronouns- Adjectives-News report- Magazine article- User's Manual for new iPod- A documentary on the big cat- Why we need to save our tigers: A dialogue- Global warming- Paragraph Writing-Arguing a case-Motivation- Problem solving.

UNIT-IV

Entertainment and Employment:- One word substitutes- Parts of speech- Gerunds and infinitives- An excerpt from a short story an excerpt from a biography- A consultant interviewing employees- Your first interview- Reality TV- Writing an essay-Correcting sentences- Integrity Sense of humor.

UNIT-V

Work and Business:- A list of 100 difficult words- Articles, Quantifiers- Punctuation - Open letter to the Prime Minister Business dilemmas: An email exchange- A review of *IPL: The Inside Story*, Mark Zuckerberg: World's Youngest Billionaire- A conversation about a business idea- Pair work: Setting up a new business- Recession- Formal letters-Emails- Reports- Professionalism-Ethics, Fill in the blanks.

Text Book:

1. *Life through Language: A Holistic Approach to Language Learning*. Board of Editors, Pearson Publishers, India. 2013.

Reference Books:

1. Basic Vocabulary. Edgar Thorpe, Showick Thorpe. Pearson P. 2008.
2. Quick Solutions to Common Errors in English, Angela Bunt. MacMillan P. 2008.
3. Know Your English (Volume 1&2), by Dr. S. Upendra, Universities Press, India 2012
4. Business Communication Strategies. Mathukutty Monippally. Tata Mc Grahill P. 2009.

MATHEMATICS - I
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that is essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Learn about linear first order and higher order ordinary differential equations and their applications.
4. Acquire knowledge of Laplace transform, partial differentiation and their applications.
5. Learn certain first order and higher order partial differential equations.

Course Outcomes: At the end of the Course, Student will be able to:

1. Solve linear ordinary differential equations of first order and first degree. Also will be able to apply the knowledge in simple applications such as Newton's law of cooling, orthogonal trajectories and simple electrical circuits.
2. Solve linear ordinary differential equations of second order and higher order. Also will be able to apply the knowledge in simple applications such as LCR circuits and Simple harmonic motion.
3. Determine Laplace transform and inverse Laplace transform of various functions.
4. Use Laplace transforms to solve a linear ODE.
5. Calculate total derivative, Jacobian and maxima/minima of functions of two variables.
6. Form partial differential equations and solve some standard types of first order PDEs. Find complimentary function and particular integral of linear higher order homogeneous and non-homogeneous PDEs.

SYLLABUS

UNIT I: Differential equations of first order and first degree:

Linear, Bernoulli, Exact, Reducible to exact types.

Applications: Newton's Law of cooling, Law of natural growth and decay, Orthogonal trajectories, Simple electrical circuits, Chemical reactions.

UNIT II: Linear differential equations of higher order:

Non-homogeneous equations of higher order with constant coefficients with RHS term of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax}V(x)$, $xV(x)$, Method of Variation of parameters.

Applications: LCR circuit, Simple Harmonic motion.

UNIT III: Laplace transforms:

Laplace transforms of standard functions, transforms of $tf(t)$, $f(t)/t$, properties, transforms of derivatives and integrals, transforms of unit step function, Dirac delta function, Inverse Laplace transforms, convolution theorem (without proof).

Applications: Solving ordinary differential equations (initial value problems) using Laplace transforms.

UNIT IV: Partial differentiation:

Introduction, Homogeneous functions, Euler's theorem, Total derivative, Chain rule, which variable is to be treated as constant, Functional dependence, Jacobians, Taylor series for a function of two variables, Leibnitz rules for differentiation under the integral sign.

Applications: Errors and Approximations, Maxima and Minima of functions of two variables without constraints, Lagrange's method (with constraints)

UNIT V: First order and higher order partial differential equations:

Formation of partial differential equations by elimination of arbitrary constants and arbitrary functions, solutions of first order Lagrange linear equation and nonlinear equations of standard types (excluding Charpit's method).

Solutions of Linear homogeneous and non-homogeneous Partial differential equations with constant coefficients - RHS terms of the type e^{ax+by} , $\sin(ax+by)$, $\cos(ax+by)$, $x^m y^n$.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
2. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson.
3. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
4. Peter O'Neil, Advanced Engineering Mathematics, Cengage Learning.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING CHEMISTRY
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Knowledge of basic concepts of chemistry for engineering students will help them as professional engineers later in design and material selection as well as utilizing the available resources.

Course Objectives:

1. To understand the physical and mechanical properties of Polymers/Plastics/elastomers helps in selecting suitable materials for different purpose.
2. To create awareness on fuels as a source of energy for industries like thermal power stations, steel industry, fertilizer industry etc.
3. To understand the concepts of galvanic cells and corrosion with theories like electro chemical theory.
4. To understand the importance of water.
5. To understand about the materials which are used in major industries like steel and metallurgical manufacturing industries, construction and electrical equipment manufacturing industries.

Course Outcomes:

1. At the end of the course the students learn the advantages and limitations of plastic materials and their use in design.
2. Fuels which are used commonly and their economics, advantages and limitations are discussed.
3. Students gained Knowledge reasons for corrosion and some methods of corrosion control.
4. Students understands the impurities present in raw water, problems associated with them and how to avoid them.
5. Similarly students understand liquid crystals and semi conductors. Students can gain the building materials , solar materials, lubricants and energy storage devices.

SYLLABUS

UNIT-I: High Polymers and Plastics; Rubbers & Elastomers:

Polymerization Definition, Types of Polymerization, Mechanism of addition polymerization, Plastics as engineering materials, Thermoplastics and Thermosetting plastics, Compounding of plastics, Fabrication of plastics (4 techniques); Preparation, Properties and applications of Polyethylene, PVC, Bakelite, Nylon - 6,6, Bullet Proof plastics -polycarbonate and Kelvar; Fiber reinforced plastics, conducting polymers, Biodegradable Polymers - PHBV, Nylon 2, Nylon 6. Natural rubber – Vulcanization – Compounding of Rubber; Preparation, properties and applications of Buna – S; Buna – N;

UNIT-II :Fuel Technology& Lubricants:

Fuels: - Introduction – Classification of fuels, Calorific value – HCV and LCV, Determination of Calorific value by bomb calorimeter; Proximate and ultimate analysis of coal, coke: manufacture of coke by Otto – Hoffmann’s by-product coke oven process; Refining of Petroleum, Knocking-octane number of gasoline, cetane number of diesel oil. Synthetic Petrol; LPG , CNG.

Lubricants:- Definition, Mechanism of Lubrication, Properties of Lubricants (Definition and significance)

UNIT-III: Electrochemical cells and Corrosion:

Galvanic cell, single electrode potential, Calomel electrode; Modern batteries: - Lead – Acid battery; Fuel cells- Hydrogen – Oxygen cell, Lithium battery Theories of corrosion (i) dry Corrosion (ii) wet corrosion. Types of corrosion - differential aeration corrosion, pitting corrosion, galvanic corrosion, stress corrosion, Factors influencing corrosion, Protection from corrosion- material selection & design, cathodic protection, Protective coatings- metallic coatings – Galvanizing, Tinning, Electroplating; Electroless plating ; Paints.

UNIT-IV: Water technology:

Sources of water – Hardness of water – Estimation of hardness of water by EDTA method; Boiler troubles – sludge and scale formation, Boiler corrosion, caustic embrittlement, Priming and foaming; Softening of water by Lime – Soda Process, Zeolite Process, Ion – Exchange Process; Municipal water treatment; Desalination of sea water by Electrodialysis and Reverse osmosis methods.

UNIT-V: Chemistry of Engineering Materials& Advanced Engineering materials

Cement:- Manufacture of Portland cement, setting and hardening of cement, Deterioration of cement concrete.

Refractories: - Definition, Characteristics, classification, Properties and failure of refractories.

Solar Energy: - Construction and working of Photovoltaic cell, applications.

Solid State Materials: Crystal imperfections, Semi Conductors, Classification and chemistry of semi conductors: Intrinsic semiconductors; Extrinsic semiconductors; Defect semiconductors, Compound Semiconductors and Organic Semiconductors.

Liquid Crystals: - Definition – Classification with examples – Applications

Text Books:

1. Engineering Chemistry by Jain and Jain, Dhanpat Rai publishing co.
2. Engineering Chemistry by Willy India Pvt Ltd.
3. Engineering Chemistry by Dr.K.Anji Reddy and Dr.M.S.R.Reddy; Silicon Publications.

Reference Books:

1. Engineering Chemistry by Shikha Agarwal; Cambridge University Press,2015 edition.
2. A text of Engineering Chemistry by S.S.Dara; S.Chand & Co Ltd.

ENGINEERING MECHANICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To impart the basic concepts of Engineering Mechanics and the principles of various force systems under static and dynamic conditions.
2. Comprehensive, theory based understanding of the underpinning natural and physical sciences and the engineering fundamentals applicable to the engineering discipline.
3. Application of established engineering methods to complex engineering problem solving.
4. The course is the key prerequisite course to sequences of courses dealing with mechanics of machines, stress analysis and design of mechanical systems.

Course Outcomes:

Students will be able to

1. Determine the resultant of the given force systems.
2. Analyze force systems using equations of equilibrium.
3. Determine centroid, center of gravity and moment of inertia of areas and bodies.
4. Analyze trusses and simple beams.
5. Distinguish between kinematics and kinetics.
6. Apply the work energy and impulse momentum methods of various engineering problems.

SYLLABUS

UNIT-I

Basic Concepts:

Scalar and vector quantities- Representation of vectors- Free vector force, Specification of force- Effect of force on rigid body- Free body diagram. Concurrent Forces in a plane: Principles of statics-Resolution and Composition of forces in a plane-Equilibrium of concurrent forces in a plane- Method of projections- Equilibrium of three forces in a plane Method of moments. Parallel Force system in a plane.

UNIT-II

Centroid & Moment of Inertia: Centroid & M.I – Area & Mass M.I – Radius of Gyration, Parallel axis– Perpendicular axis theorem – Simple Problems.

UNIT-III

General Case of Forces in a Plane: Resultant and equilibrium of general case of forces in a plane, Statically determinate plane trusses-Method of joints and Method of sections.

Friction – Coulombs laws of dry friction – Limiting friction, Problems on Wedge friction, Belt Friction-problems.

UNIT-IV

Dynamics of Particles - Rectilinear Motion – Kinematics, D'Alembert's principle, Kinetics – Work & Energy – Impulse Moment, Direct Central Impact – coefficient of restitution.

Curvilinear Motion – Kinematics, Projectile Motion, Moment of momentum, Work & Energy in Curvilinear motion.

UNIT-V

Dynamics of Rigid Bodies - Rigid body rotation – Kinematics - Kinetics – Work & Energy in Rigid body rotation, Plane Motion – Kinematics – Instantaneous center of rotation, Kinetics - Work-Energy principle in plane motion.

Text Book:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.

Reference Books:

1. Engineering Mechanics, Vol.1 & 2 by J.L. Meriems and L.G. Kraige.
2. Engineering Mechanics by Singer.
3. Engineering Mechanics by K.L. Kumar, Tata Mc-Graw Hill.
4. Engineering mechanics by Bhavikatti, New age international.

ENGINEERING DRAWING
(Common to CIV, EEE & ME)

Lecture	: 1 Period	Int.Marks	: 30
Practice	: 3 Periods	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes:

Students will be able to

1. Apply principles of drawing to represent dimensions of an object.
2. Construct polygons and engineering curves.
3. Draw projections of points, lines, planes and solids.
4. Represent the object in 3D view through isometric views.
5. Convert the isometric view to orthographic view and vice versa.

SYLLABUS

UNIT I

Polygons: Constructing regular polygons by general methods, inscribing and describing polygons on circles.

Curves: Parabola, Ellipse and Hyperbola by general methods, cycloids, involutes, tangents & normals for the curves.

UNIT II

Orthographic Projections: Horizontal plane, vertical plane, profile plane, importance of reference lines, projections of points in various quadrants, projections of lines, lines parallel either to one of the reference planes (HP, VP or PP)

Projections of straight lines inclined to both the planes, determination of true lengths, angle of inclination and traces- HT, VT

UNIT III

Projections of planes: regular planes perpendicular/parallel to one plane and inclined to the other reference plane; inclined to both the reference planes.

UNIT IV

Projections of Solids – Prisms, Pyramids, Cones and Cylinders with the axis inclined to one of the planes.

UNIT V

Conversion of isometric views to orthographic views; Conversion of orthographic views to isometric views.

Text Books:

1. Engineering Drawing by N.D. Butt, Chariot Publications.
2. Engineering Drawing by Agarwal & Agarwal, Tata McGraw Hill Publishers

Reference Books:

1. Engineering Drawing by K.L.Narayana & P. Kannaiah, Scitech Publishers.
2. Engineering Graphics for Degree by K.C. John, PHI Publishers.
3. Engineering Graphics by PI Varghese, McGrawHill Publishers.
4. Engineering Drawing + AutoCad – K Venugopal, V. Prabhu Raja, New Age

ENVIRONMENTAL STUDIES
(Common to all Branches)

Lecture : 2 Periods

Tutorial : 1 Period.

Exam : 3 Hrs.

Int.Marks : 30

Ext. Marks : 70

Credits : 2

Course Objectives:

1. Developing an awareness and sensitivity to the total environment and its related problems.
2. Motivating students for active participation in environmental protection and improvement.
3. Developing skills for active identification and development of solutions to environmental problems
4. Evaluation of environment programmes in terms of social, economic, ecological and aesthetic factors.
5. Creating a “CONCERN AND RESPECT FOR THE ENVIRONMENT”.

Course Outcomes:

1. To bring awareness among the students about the nature and natural ecosystems
2. Sustainable utilization of natural resources like water, land, energy and air
3. Resource pollution and over exploitation of land, water, air and catastrophic (events) impacts of climate change, global warming, ozone layer depletion, marine, radioactive pollution etc to inculcate the students about environmental awareness and safe transfer of our mother earth and its natural resources to the next generation
4. Safe guard against industrial accidents particularly nuclear accidents
5. Constitutional provisions for the protection of natural resources

SYLLABUS

UNIT – I

Global Environmental Crisis:

Environmental Studies - Definition, Scope and importance, Need for public awareness. Global Environmental Crisis

Ecosystems:

Basic Concepts - Structure and Functions of Ecosystems: Producers, Consumers and Decomposers. Types of Ecosystems: Forest Ecosystems, Grassland Ecosystems Desert Ecosystems and Aquatic Ecosystems

UNIT-II

Biodiversity:

Introduction to Biodiversity, Values of Bio-diversity, Bio-geographical classification of India, India as a Mega-diversity habitat, Threats to biodiversity, Hotspots of Biodiversity, Conservation of Biodiversity: In-situ and Ex-situ conservation of Biodiversity.

UNIT-III

Environmental and Natural Resources Management:

Land Resources: Land degradation, soil erosion and desertification, Effects of modern agriculture. **Forest Resources:** Use and over exploitation-Mining and Dams-their effects on forest and tribal people. **Water resources:** Use and over utilization of surface and ground water, Floods, droughts, conflict over water, water logging and salinity, dams – benefits and problems. **Energy Resources:** Renewable and non-renewable energy sources, use of alternate energy sources-impact of energy use on environment.

UNIT-IV

Environmental Pollution:

Causes, Effects and Control measures of - Air pollution, Water pollution, Soil pollution, Marine Pollution, Thermal pollution, Noise pollution, Nuclear Hazards; Climate change and Global warming, Acid rain and Ozone layer depletion. Solid Waste Management: Composting, Vermiculture, Urban and Industrial Wastes, Recycling and Reuse.

Environmental Problems in India:

Drinking water, Sanitation and Public health, Population growth and Environment; Water Scarcity and Ground Water Depletion; Rain water harvesting, Cloud seeding and Watershed management.

UNIT-V

Institutions and Governance:

Regulations by Government- Environmental Protection Act, Air (Prevention & Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act. Environmental Impact Assessment (EIA)

Case Studies:

Chipko Movement, Narmada Bachao Andolan, Silent Valley Project, Mathura Refinery and Taj Mahal, Industrialization of Patancheru, Nuclear reactor at Nagarjuna Sagar, Tehri Dam, Ralegaon Siddhi (Anna Hazare), Kolleru lake – Aquaculture, Fluorosis in Andhra Pradesh & Telangana.

Field Work:

Visit to a local area to document and mapping environmental assets. Visits to Industries, Water Treatment Plants, Effluent Treatment Plants.

Text Books:

1. Environmental Studies, K. V. S. G. Murali Krishna, VGS Publishers, Vijayawada
2. Environmental Studies, R. Rajagopalan, 2nd Edition, 2011, Oxford University Press.
3. Environmental Studies, Dr. M. Sita Rama Reddy, Dr. K. Anji Reddy, Silicon Publications, ASR Nagar, Bhimavaram
4. Environmental Studies, P. N. Palanisamy, P. Manikandan, A. Geetha, and K. Manjula Rani; Pearson Education, Chennai

Reference Books:

1. Text Book of Environmental Studies, Deeshita Dave & P. Udaya Bhaskar, Cengage Learning.
2. A Textbook of Environmental Studies, Shaashi Chawla, TMH, New Delhi
3. Environmental Studies, Benny Joseph, Tata McGraw Hill Co, New Delhi
4. Perspectives in Environment Studies, Anubha Kaushik, C P Kaushik, New Age International Publishers, 2014

**ENGINEERING CHEMISTRY LAB
(Common to CIV, EEE & ME)****Lab : 3 Periods**
Exam : 3 Hrs.**Int.Marks : 50**
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To investigate and understand Physical behavior in the laboratory using scientific reasoning and logic and interpret the result of simple experiments and demonstration of chemical Principle and also evaluate the impact of chemical discoveries on how we view the world.
2. Effectively communicate experimental results and solutions to application problems through oral and written reports.
3. Understand the basic concepts, definitions, characteristics and phenomena.
4. Recognize the classical ideas and chemical phenomena and also define and analyse the concepts.

Course Outcomes:

1. An understanding of Professional and develop confidence on recent trends.
2. Able to gain technical knowledge of measuring, operating and testing of chemical instruments and equipments.
3. Acquire ability to apply knowledge of chemistry.
4. Exposed to the real time working environment.
5. Demonstrate the ability to learn Principles, design and conduct experiments.
6. Ability to work on laboratory and multidisciplinary tasks.

LIST OF EXPERIMENTS**Introduction to chemistry Laboratory**

1. Estimation of HCl using standard Sodium Hydroxide.
2. Determination of total hardness of water by EDTA method.
3. Estimation of Ferrous Iron by KMnO_4 .
4. Estimation of oxalic acid by KMnO_4
5. Estimation of Mohr's salt by $\text{K}_2\text{Cr}_2\text{O}_7$
6. Estimation of Dissolved oxygen by Winkler's method.
7. Determination of pH by pH meter and universal indicator method.
8. Conductometric titration of strong acid Vs strong base
9. Conductometric titration of strong acid Vs weak base.
10. Potentiometric titration of strong acid Vs strong base
11. Potentiometric titration of strong acid Vs weak base
12. Preparation of Phenol formaldehyde resin.
13. Determination of saponification value of oils
14. Determination of pour and cloud points of lubricating oil.
15. Determination Acid value of oil.

Text Books:

1. Engineering Chemistry Lab Manual Prepared by Chemistry Faculty of S.R.K.R.Engineering College.

Reference Books:

1. Laboratory manual on Engineering Chemistry by Dr.Sudha Rani ;Dhanpat Rai Publishing Company.
2. Engineering Chemistry Laboratory manual – I & II by Dr. K.Anji Reddy; Tulip Publications.

ENGLISH COMMUNICATION SKILLS LAB- I
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn through practice the communication skills of listening, speaking, reading and writing.
2. To make students recognize the sounds of English through Audio- Visual aids.
3. To familiarize the students with stress and intonation.
4. To help the students build their confidence in speaking skills.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students improve their speaking skills in real contexts.
3. Students learn standard pronunciation and practice it daily discourse.
4. Students give up their communicative barriers.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

ENGINEERING WORKSHOP & IT WORKSHOP
(Common to CIV, EEE & ME)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

PART-A ENGINEERING WORKSHOP

Course Objective:

To impart hands-on practice on basic engineering trades and skills.

Course Outcomes:

Student will be able to

1. Use various tools to prepare basic carpentry and fitting joints.
2. Prepare jobs of various shapes using black smithy.
3. Make basic house wire connections.
4. Fabricate simple components using tin smithy.

SYLLABUS

Carpentry	Fitting
1. T-Lap Joint 2. Cross Lap Joint 3. Dovetail Joint 4. Mortise and Tenon Joint	1. Vee Fit 2. Square Fit 3. Half Round Fit 4. Dovetail Fit
Black Smithy	Tin Smithy
1. Round rod to Square 2. S-Hook 3. Round Rod to Flat Ring 4. Round Rod to Square headed bolt	1. Taper Tray 2. Square Box without lid 3. Open Scoop 4. Funnel
House Wiring	
1. Parallel / Series Connection of three bulbs 2. Stair Case wiring 3. Florescent Lamp Fitting 4. Measurement of Earth Resistance	

Note: At least two exercises to be done from each trade.

Reference:

1. Elements of workshop technology, Vol.1 by S. K. and H. K. Choudary.

PART B: IT WORKSHOP:

Course Objectives:

1. Understand the basic components and peripherals of a computer.
2. To become familiar in configuring a system.
3. Learn the usage of productivity tools.
4. Acquire knowledge about the netiquette and cyber hygiene.
5. Get hands on experience in trouble shooting a system

Course Outcomes:

At the end of the course the students can able to

1. Assemble and disassemble the systems
2. Use the Microsoft office tools
3. Install various software
4. Know about various search engines
5. Trouble shoot various Hardware and Software problems
6. Use MATLAB and LATEX softwares

LIST OF EXERCISES

1. System Assembling, Disassembling and identification of Parts / Peripherals
2. Operating System Installation-Install Operating Systems like Windows, Linux along with necessary Device Drivers.
3. MS-Office / Open Office
 - a) Word - Formatting, Page Borders, Reviewing, Equations, symbols.
 - b) Spread Sheet - organize data, usage of formula, graphs, charts.
 - c) Power point - features of power point, guidelines for preparing an effective presentation.
 - d) Access- creation of database, validate data.
4. Network Configuration & Software Installation-Configuring TCP/IP, proxy and firewall settings. Installing application software, system software & tools.
5. Internet and World Wide Web-Search Engines, Types of search engines, netiquette, cyber hygiene.
6. Trouble Shooting-Hardware trouble shooting, Software trouble shooting.
7. MATLAB- basic commands, subroutines, graph plotting.
8. LATEX-basic formatting, handling equations and images.

Reference Books:

1. Computer Hardware, Installation, Interfacing, Troubleshooting and Maintenance, K.L. James, Eastern Economy Edition.
2. Microsoft Office 2007: Introductory Concepts and Techniques, Windows XP Edition By Gary B. Shelly, Misty E. Vermaat and Thomas J. Cashman (2007, Paperback).
3. LATEX- User's Guide and Reference manual, Leslie Lamport, Pearson, LPE, 2/e.
4. Getting Started with MATLAB: A Quick Introduction for Scientists and engineers, Rudraprathap, Oxford University Press, 2002.
5. Scott Mueller's Upgrading and Repairing PCs, 18/e, Scott. Mueller, QUE, Pearson, 2008
6. The Complete Computer upgrade and repair book, 3/e, Cheryl A Schmidt, Dreamtech.
7. Comdex Information Technology course tool kit Vikas Gupta, WILEY Dreamtech.
8. Introduction to Information Technology, ITL Education Solutions limited, Pearson Education.

**INNER ENGINEERING
(Common to CIV, EEE & ME)**

Practice : 2 Periods

Course Objectives:

Student should learn:

1. Human values and tools to lead a happy, stress-free life.
2. Yoga asanas, Pranayama, Sudarshan Kriya & Meditation
3. At-least two creative arts out of photography, sketching, craft-making, singing, clay molding, upcycling etc.
4. Concentration pranayama, Ego bursting process.
5. To take up responsibility for society and teach classes of their choice to school children.
6. About good food habits for good health.

Course Outcomes:

Student should be capable of

1. To improve his concentration levels and improve his public speaking abilities.
2. To balance his academic and non-academic activities (Work Life Balance).
3. To widen his vision and increase his breadth of perspective in his journey of 4 years.
4. To improve his communications skills, leadership, teamwork and decision-making abilities.
5. To inculcate creativity & innovation, planning & organizing as part of their life.
6. Taking responsibility for themselves and people around them.
7. To make their journey more fun and enjoyable.

SYLLABUS

UNIT-I

YES!+ Workshop:

Yoga Postures – Seven Layers To our Existence – Puzzles – Sources Of Energy – Live in the present Moment – Importance of Breath – Ujjai Breath – Pranayama – Sudarshana Kriya.

UNIT-II

YES!+ Workshop:

Yoga Postures (Suryanamaskars) – Giving 100% in everything – Time management – Happiness point – Opposite Values – Pranayama – Sudarshan kriya

UNIT-III

YES!+ Workshop:

Yoga Postures – Knowledge points (Acceptance, opinions discretion and handling mistakes) – Eye Gazing Process – Dance – Life Story process – Sudarshana Kriya (short) – Eternal life – Ego Bursting – Relationships – Parents – Studies – Compliments/Praising process.

UNIT-IV

Creative Arts:

Photography – Sketching – Handy-crafts – Clay molding – Singing – Upcycling – Communing with nature – Creative writing.

UNIT -V

Service:

Leadership in action – Contributing to society – Take up Responsibility –Empowerment – Public Speaking – Art of Teaching.

Reference Books:

1. Discourse on Patanjali Yoga Sutras by H.H Sri SriRavishankar
2. Human values and professional ethics byRRGaur,RSangal,GPBagaria
3. The Art of Photography by AL Judge
4. Hand made in India : Crafts of India by Ranjan Aditi

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)
I/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)
Under Choice Based Credit System

GROUP-A (CIVIL, EEE & MECHANICAL)
II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/ Week	Internal Marks	External Marks	Total Marks
B17 BS 1201	English – II *	3	3	1	--	4	30	70	100
B17 BS 1202	Mathematics – II	3	3	1	--	4	30	70	100
B17 BS 1203	Mathematics – III*	3	3	1	--	4	30	70	100
B17 BS 1204	Engineering Physics	3	3	1	--	4	30	70	100
B17 CS 1201	Computer Programming Using C	3	3	1	--	4	30	70	100
# DS 1	Department Subject	3	3	1	--	4	30	70	100
B17 BS 1206	Engineering Physics Lab	2	--	--	3	3	50	50	100
B17 BS 1208	English Communication Skills Lab – II *	2	--	--	3	3	50	50	100
B17 CS 1204	C Programming Lab	2	--	--	3	3	50	50	100
B17 BS 1210	Engineering Physics Virtual Labs- Assignments	--	--	--	2	2	--	--	--
B17 BS 1211	NCC	--	--	--	2	2	--	--	--
Total		24	18	6	13	37	330	570	900

*** Common to both Group - A and Group - B**

#DS 1	CIVIL	B17 CE 1201	Building Materials and Construction
	EEE	B17 EE 1201	Circuit Theory
	MECHANICAL	B17 EE 1202	Basic Electrical and Electronics Engineering

ENGLISH - II
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To improve the language proficiency of the students in English with emphasis on LSRW skills.
2. To enable the students to study and comprehend the prescribed lessons and subjects more effectively relating to their theoretical and practical components.
3. To develop spoken and written forms of the students in both formal and informal situations.
4. To enrich the vocabulary of the students through the usage of the lexis in appropriate contexts.
5. To expose them to the correct structural patterns of English.

Course Outcomes: At the end of the Course, Student will be able to:

1. To comprehend the speech of people belonging to different backgrounds and regions.
2. Understand the importance of speaking and writing for personal and professional communication and practice it in real contexts.
3. To express fluently and accurately in social discourse.
4. Participate in group activities like role-plays, discussions and debates.
5. Identify the discourse features, and improve intensive and extensive reading skills.

SYLLABUS

UNIT I:

- A. Detailed-Text: Unit 1: ' The Greatest Resource- Education'
- B. Non-Detailed Text: Lesson 1: ' A P J Abdul Kalam' from The Great Indian Scientists.

UNIT II:

- A. Detailed-Text: Unit 2: ' A Dilemma'
- B. Non-Detailed Text: Lesson 2:'C V Raman' from The Great Indian Scientists.

UNIT III:

- A. Detailed-Text: Unit 3: 'Cultural Shock': Adjustments to new Cultural Environments
- B. Non-Detailed Text: Lesson 3:'Homi Jehangir Bhabha' from The Great Indian Scientists.

UNIT IV:

- A. Detailed-Text: Unit 4: 'The Lottery'
- B. Non-Detailed Text: Lesson 4: 'Jagadish Chandra Bose' from The Great Indian Scientists.

UNIT V:

- A. Detailed-Text: Unit 5: ' The Chief Software Architect'
- B. Non-Detailed Text: Lesson 5: ' Prafulla Chandra Ray' from The Great Indian Scientists

Detailed Textbook:

1. English Encounters Published By Maruthi Publishers.

Non-Detailed Text Book:

1. The Great Indian Scientists Published by Cengage learning.

MATHEMATICS – II
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand some basic numerical methods to solve algebraic and transcendental equations.
4. Learn interpolation methods for equally spaced and unequally spaced data points. Learn methods for numerical evaluation of integrals and for solving first order ODEs.
5. Learn concepts of a Fourier series and Fourier transform.

Course Outcomes: At the end of the Course, Student will be able to:

1. Find a real root of algebraic and transcendental equations using different methods.
2. Know the relation between the finite difference operators. Determine interpolation polynomial for a given data.
3. Evaluate numerically certain definite integrals applying Trapezoidal and Simpson's rules.
4. Solve a first order ordinary differential equation by Euler and RK methods.
5. Find Fourier series of a given function satisfying Dirichlet conditions. Find half range cosine and sine series for appropriate functions.
6. Find Fourier transforms Fourier cosine and sine transforms of appropriate functions and evaluate certain integrals using inverse transforms and Fourier integral.

SYLLABUS

UNIT I: Solution of Algebraic and Transcendental Equations:

Introduction, Bisection method, Method of false position, Iteration method, Newton- Raphson method (One variable and simultaneous Equations).

UNIT II: Interpolation:

Introduction, Errors in polynomial interpolation, Finite differences, Forward differences, Backward differences, Central differences and Symbolic relations between the operators, Differences of a polynomial, Newton's formulae for interpolation, Interpolation with unequal intervals, Lagrange's interpolation formula.

UNIT III: Numerical Integration and solution of Ordinary Differential equations:

Trapezoidal rule, Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rules, Solution of ordinary differential equations by Taylor series method, Picard's method of successive approximations, Euler's method, Runge-Kutta methods (second order and fourth order).

UNIT IV: Fourier Series:

Introduction, Periodic functions, Fourier series of a periodic function, Dirichlet's conditions, Even and odd functions, Change of interval, Half-range sine and cosine series, Parseval's formula.

UNIT V: Fourier Transforms:

Fourier integral theorem (without proof), Complex form of Fourier integral, Fourier sine and cosine integrals, Fourier transform, Fourier sine and cosine transforms, properties, inverse transforms, Parseval's identities, Finite Fourier transforms.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, A Text book of Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Dean G. Duffy, Advanced engineering mathematics with MATLAB, CRC Press.
2. V.Ravindranath and P. Vijayalakshmi, Mathematical Methods, Himalaya Publishing House.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
4. David Kincaid, Ward Cheney, Numerical Analysis-Mathematics of Scientific Computing, 3rd Edition, Universities Press.
5. Srimanta Pal, Subodh C. Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

MATHEMATICS - III
(Common to all Branches)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The course is designed to equip the students with the necessary mathematical skills and techniques that are essential for an engineering course.
2. The skills derived from the course will help the student form a necessary base to develop analytic and design concepts.
3. Understand important concepts of Linear algebra including solution of linear simultaneous equations, eigen values and eigen vectors.
4. Learn Beta, Gamma functions and how to evaluate double and triple integrals.
5. Learn fundamental concepts of vector calculus that help understand other Engineering courses.

Course Outcomes: At the end of the Course, student will be able to:

1. Determine rank, and solve a system of linear simultaneous equations numerically using various matrix methods.
2. Determine Eigen values and Eigen vectors of a given matrix, Reduce a Quadratic form to its canonical form and classify.
3. Evaluate double integrals over a region and triple integral over a volume.
4. Use the knowledge of Beta and Gamma functions in evaluation of different integrals.
5. Find gradient of a scalar function, divergence and curl of a vector function. Use vector identities for solving problems.
6. Evaluate line, surface and volume integrals by the use of Green's, Stokes' and Gauss divergence theorems.

SYLLABUS

UNIT I: Linear systems of equations:

Rank, Echelon form, Normal form, Solution of linear systems, Gauss elimination, Gauss-Jordan, Jacobi and Gauss-Seidel methods.

Applications: Finding the current in electrical circuits.

UNIT II: Eigen values - Eigen vectors and Quadratic forms:

Eigen values, Eigen vectors, Properties, Cayley-Hamilton theorem, Inverse and powers of a matrix by using Cayley-Hamilton theorem, Diagonalization, Quadratic forms, Reduction of a Quadratic form to Canonical form, Rank, Positive, Negative, Semi-Definite and indefinite forms of a Quadratic form, Index and Signature of a Quadratic form.

Applications: Free vibration of a two-mass system.

UNIT III: Multiple integrals:

Double and triple integrals, Change of variables, Change of order of integration. Application to finding Areas, Moment of Inertia and Volumes.

Beta and Gamma functions, Properties, Relation between Beta and Gamma functions, Application to evaluation of improper integrals. The error function and the complimentary error function.

UNIT IV: Vector Differentiation:

Gradient, directional derivative, Divergence, Curl, Incompressible flow, solenoidal and irrotational vector fields, second order operators, vector identities.

UNIT V: Vector Integration:

Line integral, Work done, Potential function; Area, Surface and volume integrals, Flux, Vector integral theorems: Greens, Stokes and Gauss Divergence theorems (without proof) and related problems.

Text Books:

1. B.S.Grewal, Higher Engineering Mathematics, 43rd Edition, Khanna Publishers.
2. N.P.Bali & Manish Goyal, Engineering Mathematics, Lakshmi Publications.

Reference Books:

1. Michael Greenberg, Advanced Engineering Mathematics, 9th edition, Pearson edn.
2. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley-India.
3. Peter O'Neil, Advanced Engineering Mathematics, 7th edition, Cengage Learning.
4. D.W. Jordan and T. Smith, Mathematical Techniques, Oxford University Press.
5. Srimanta Pal, Subodh C.Bhunia, Engineering Mathematics, Oxford University Press.
6. Dass H.K., Rajnish Verma. Er., Higher Engineering Mathematics, S. Chand Co. Pvt. Ltd, New Delhi.

ENGINEERING PHYSICS
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

To re-orient Physics curriculum to the needs of the circuit and non circuit branches of Engineering / Technology courses offered by SRKREC(A) Bhimavaram that serves to understand the general and branch specific basic topics and introduce the related advanced technologies to the stake holders.

1. To impart the knowledge of physical optics phenomena like Interference and Diffraction required to understand the design and working of optical instruments with high resolution.
2. To understand the concept of coherence and generation of high intensity light sources and propagation of light waves in fibers for technological applications.
3. To study and analyze the behavior of electric and magnetic fields and their mutual interaction, and to understand the science of ultrasonics.
4. To understand the basic concepts of Quantum mechanics to know the behavior of fundamental particles in materials and to understand the basic nature of materials in general and classify them.
5. To study the structure property relationship of crystalline solids and understand basics of nanotechnology.

Course Outcomes: At the end of the course the students-

1. Learn the basic concepts of interference and diffraction of light and its applications.
2. Understand the science of producing high intensity light beams for technological applications and also understand the propagation of light waves in optical fibers in various applications.
3. Understand the inter relationship of electric and magnetic fields and learn ultrasonic's as a tool for technological applications
4. Learn the behavior of particles at the very microscopic level by using wave nature of particles and understand the behavior of materials and be able to classify them using the band theory of solids.
5. Learn the basics of structures of solid materials and nano material preparation Techniques/methods.

SYLLABUS

UNIT I: Interference and Diffraction

Principle of superposition-coherence-interference in thin films (reflected system) – Wedge shaped film-Newton's rings-Michelson's interferometer. Fraunhofer's diffraction at single slit, Diffraction grating-Resolving power of a grating.

UNIT- II: Lasers and Optical Fibers

Introduction, Spontaneous emission and Stimulated emission – Einstein's relation – Requirements of Laser device- Ruby laser- He-Ne gas laser- Characteristics of laser- Applications.

Description of optical fiber, Principle of light propagation- Optical fiber –Acceptance angle- Numerical aperture of optical fiber- Modes of propagation- Classification of fibers- Applications of fiber.

UNIT- III: Electro Magnetic Fields and Ultrasonics

Concept of Electromagnetic induction , Faraday's law, Lenz's law, Electric fields due to time varying magnetic fields, Magnetic fields due to time varying electric fields, Displacement current, Modified Ampere's law, Maxwell's equations and their significance (without derivation).

Definition of Ultrasonics-Methods of Producing Ultrasonics- Detection of Ultrasonics- Applications of Ultrasonics.

UNIT- IV: Quantum Mechanics and Band Theory of Solids

Introduction, de Broglie matter waves- properties-Experimental confirmation, wave function-significance- Schrodinger's time dependent and time independent wave equations- Eigen values and functions, Particle in a box.

Band theory of Solids- Introduction- Kronig Penney model (Qualitative)- Energy bands of crystalline solids- Distinction between Conductors, Semi conductors and insulators.

UNIT-V: Crystallography and Nano Materials

Basis and Lattice, Crystal systems, Bravais lattice, Unit cell Coordination number – Packing fraction for SC ,FCC, and BCC lattices, Miller indices- Diffraction of X rays from crystals- Bragg's law.

Introduction to Nanomaterials – Synthesis methods : Condensation, ball milling, sol-gel, chemical vapour deposition methods, properties and applications.

Text Books:

1. Physics by Resnick & Halliday. Wiley - Eastern (India) Ltd.
2. Engineering Physics By M.N. Avadhanulu & PG Kshirasagar, S.Chand & Co.
3. Engineering Physics by V. Rajendran, Mc Graw Hill Education (India) Pvt.Ltd.

Reference Books:

1. Engineering Physics by MR Srinivasan, New International Publishers.
2. Solid State Physics – A.J. Dekkar, MacMillan (India) Ltd.
3. Engineering Physics by Gaur Gupta, Dhanpat Rai Publications, Meerut, India.
4. Engineering Physics by P k Palanisamy, Scitech Publications (India) Pvt.Ltd

(Note: Assignment Marks of **Engineering Physics** are to be considered from the Internal marks of **Engineering Physics-- Virtual Labs – Assignments B17 BS 1210**)

COMPUTER PROGRAMMING USING C
(Common to CIV, EEE & ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Formulating algorithmic solutions to problems and implementing algorithms in C

1. Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux
2. understanding branching, iteration and data representation using arrays
3. Modular programming and recursive solution formulation
4. Understanding pointers and dynamic memory allocation
5. Understanding miscellaneous aspects of C
6. Comprehension of file operations

Course Outcomes:

1. Understand the basic terminology used in computer programming
2. Write, compile and debug programs in C language.
3. Use different data types in a computer program.
4. Design programs involving decision structures, loops and functions.
5. Explain the difference between call by value and call by reference
6. Understand the dynamics of memory by the use of pointers
7. Use different data structures and create/update basic data files.

SYLLABUS

UNIT I:

Unit objective: Notion of Operation of a CPU, Notion of an algorithm and computational procedure, editing and executing programs in Linux

Introduction: Computer systems, Hardware and Software Concepts.

Problem Solving: Algorithm / Pseudo code, flowchart, program development steps, computer languages: machine, symbolic and highlevel languages, Creating and Running Programs: Writing, Editing(vi/emacs editor), Compiling(gcc), Linking and Executing in under Linux.

BASICS OF C: Structure of a c program, identifiers, basic data types and sizes. Constants, Variables, Arithmetic , relational and logical operators, increment and decrement operators, conditional operator, assignment operator, expressions, type conversions, Conditional Expressions, precedence and order of evaluation, Sample Programs.

UNIT II:

Unit objective: understanding branching, iteration and data representation using arrays

SELECTION – MAKING DECISION: TWO WAY SELECTION: if-else, null else, nested if, examples, Multi-way selection: switch, else-if, examples.

ITERATIVE: loops- while, do-while and for statements , break, continue, initialization and updating, event and counter controlled loops, Looping applications: Summation, powers, smallest and largest.

ARRAYS: Arrays- concepts, declaration, definition, accessing elements, storing elements, Strings and String Manipulations, 1-D arrays, 2-D arrays and character arrays, string manipulations, Multidimensional arrays, array applications: Matrix operations, checking the symmetricity of a Matrix. **STRINGS: concepts, c strings.**

UNIT III:

Objective: Modular programming and recursive solution formulation

FUNCTIONS- MODULAR PROGRAMMING: functions, basics, parameter passing, storage classes extern, auto, register, static, scope rules, block structure, user defined functions, standard library functions, recursive functions, Recursive solutions for fibonacci series, towers of Hanoi, header files, C Preprocessor, example c programs, Passing 1-D arrays, 2-D arrays to functions.

UNIT IV:

Objective: Understanding pointers and dynamic memory allocation

POINTERS: pointers- concepts, initialization of pointer variables, pointers and function arguments, passing by address- dangling memory, address arithmetic, character pointers and functions, pointers to pointers, pointers and multi-dimensional arrays, dynamic memory management functions, command line arguments

UNIT V:

Objective: Understanding miscellaneous aspects of C

ENUMERATED, STRUCTURE AND UNION TYPES: Derived types- structures declaration, definition and initialization of structures, accessing structures, nested structures, arrays of structures, structures and functions, pointers to structures, self referential structures, unions, typedef, bit-fields, program applications

BIT-WISE OPERATORS: logical, shift, rotation, masks.

Objective: Comprehension of file operations

FILEHANDLING: Input and output- concept of a file, text files and binary files, Formatted I/O, File I/O operations, example programs

Text Books:

1. Problem Solving and Program Design in C, Hanly, Koffman, 7thed, PERSON
2. Programming in C, Second Edition PradipDey and Manas Ghosh, OXFORD Higher Education
3. Programming in C, A practical approach Ajay Mittal PEARSON
4. The C programming Language by Dennis Richie and Brian Kernighan
5. Programming in C, B. L. Juneja, Anith Seth, Cengage Learning.

Reference Books:

1. C Programming, A Problem Solving Approach, Forouzan, Gilberg, Prasad, CENGAGE
2. Programming with C, Bichkar, Universities Press
3. Programming in C, ReemaThareja, OXFORD
4. C by Example, Noel Kalicharan, Cambridge

BUILDING MATERIALS AND CONSTRUCTION
(For CIVIL)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To promote development, production, standardization and large-scale application of cost-effective innovative building materials and construction technologies in housing and building sector.
2. To develop and promote methodologies and technologies related to different Building constructions
3. To promote new building materials and components through technical support and encouraging entrepreneurs to set up production units.
4. To provide standard scientific and technical services to professionals, construction agencies and entrepreneurs in selection, evaluation and design engineering materials.

Course Outcomes:

1. Define and classify various stones, clay products used in construction sector.
2. Find various types of wood their conversion and relevant BIS testing procedures to be carried out to ascertain the quality of building materials.
3. Identify the major ingredients of construction materials like lime, cement, aggregate and their use in the construction industry.
4. Select different materials for finishing's and various constructions pertaining to masonry works and foundations.
5. Develop the conceptual knowledge of various supports in building construction.

SYLLABUS

UNIT I: STONES, BRICKS AND CLAY PRODUCTS

Stones: Classification of stones, Properties of building stones, Stone quarrying, precautions in blasting **Bricks:** Classification of Bricks, Manufacture of Bricks, general qualities of Bricks as per IS code, tests for good bricks as per IS code, including field tests. **Clay Products:** Tiles-types, manufacturing and their uses, Earth-ware, Terra-cotta, stone ware, Porcelain.

UNIT II: WOOD, WOOD BASED PRODUCTS

Wood: cross section details of trees, their general properties, characteristics of good timber defects in timber, mechanical properties of timber, seasoning and its importance, Decay of timber, **Wood based Products:** Veneers, Plywood and its types, Manufacturing of plywood, plywood grades as per IS code, Laminated wood, merits of plywood and laminated wood, Lamin Boards, Block boards, Batten board, Particle boards

UNIT III: LIME, CEMENT & AGGREGATES

Lime: Various ingredients of lime, Constituents of lime stone, classification of lime ,
Cement: Natural and artificial cements, types of artificial cements and their uses, Wet and dry process of manufacturing ordinary Portland cement (OPC), composition of cement, Various field and Laboratory tests on OPC as per IS code, Storage of cement. **Aggregates:** Classification of aggregate – Coarse and fine aggregates, Particle shape and Texture, Specific gravity, Bulk density, porosity and Absorption, Moisture content of Aggregate – Bulking of sand, Sieve analysis.

UNIT IV: FINISHINGS, MASONRY AND FOUNDATIONS

Finishings: Paints and Varnishes: Constituents and characteristics of paints, types of paint and their uses, painting defects, causes and remedies. Constituents of varnishes, types of varnish and their uses, Pointing and Plastering. **Masonry:** Different types of Stone Masonry- Plan, Elevation, Sections of stone Masonry works- Brick Masonry- Different Types of Bonds- Plan, Elevation and section of Brick Bonds upto Two-Brick wall thickness- Partition walls- Different types of Block Masonry- Hollow concrete Blocks- FAL-G Blocks, Hollow Clay Blocks. **Foundations:** Types- strip, isolated, strap, combined footings, Raft-Mat- flat slab and Beam raft, box type raft.

UNIT V: ROOFING, FORM WORK & SCAFFOLDING

Roofing: Mangalore tiled roof, RCC roof, Madras terrace roof, Hollow tiled roof, Asbestos cement, Fibre glass, Aluminum G.I. Sheet roofing's. **Form work, Scaffolding:** form work- types of formwork, centering- scaffolding- types of scaffolding. Trusses: Types- King post and queen post trusses and their uses. Stair cases: Various types of stair cases- dog legged, quarter landing, spiral stairs etc.

Text Books:

1. Engineering Materials by Rangawala. Charotar Publications
2. Building Materials by S.K.Duggal, New Age International Publications.
3. Building Materials by B.C. Punmia, Laxmi Publications private ltd.
4. Building Construction by B.C. Punmia, Laxmi Publications (p) ltd.

Reference Books:

1. Building Materials by S.S. Bhavikatti, Vices publications House private ltd.
2. Building Construction by S.S. Bhavikatti, Vices publications House private ltd.
3. Building Materials by M.L.Gambhir, Tata McGraw Hill Publishing Co. Ltd. New Delhi.
4. Building Materials by P.C.Vergheze, PHI learning (P) ltd.
5. Building construction by P.C.Vergheze, PHI Learning (P) Ltd.

CIRCUIT THEORY (For EEE)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To study the concepts of passive elements, types of sources and various network reduction techniques.
2. To understand the applications of network topology to electrical circuits.
3. To study the concept of magnetic coupled circuits.
4. To understand the behavior of RLC networks for sinusoidal excitations and to understand the concept of resonance.
5. To understand the concepts of 3-phase circuits.

Course Outcomes:

Students are able to solve

1. Various electrical networks in presence of active and passive elements.
2. Electrical networks with network topology concepts.
3. Magnetic circuit with various dot conventions.
4. R, L, C network with sinusoidal excitation.
5. Three phase AC circuits.

SYLLABUS

UNIT-I: Introduction to Electrical Circuits:

Passive components and their V-I relations. Sources (dependent and independent) -Kirchhoff's laws, Network reduction techniques (series, parallel, series - parallel, star-to-delta and delta-to-star transformation). Source transformation technique, nodal analysis and mesh analysis.

UNIT-II: Network topology:

Definitions of Graph and Tree, Basic cutset and tieset matrices for planar networks, Loop and nodal methods of analysis of networks with dependent and independent voltage and current sources, Duality and Dual networks.

UNIT-III: Magnetic Circuits:

Basic definition of MMF, flux and reluctance. Analogy between electrical and magnetic circuits. Faraday's laws of electromagnetic induction Concept of self and mutual inductance. Dot convention-coefficient of coupling and composite magnetic circuit. Analysis of series and parallel magnetic circuits.

UNIT-IV: Single Phase A.C Systems:

Periodic waveforms (determination of rms, average value and form factor). Concept of phase angle and phase difference – Waveforms and phasor diagrams for lagging, leading networks. Complex and polar forms of representations, steady state analysis of R, L and C circuits. Power Factor and its significance - Real, Reactive, Apparent and Complex power. Node and mesh analysis of AC networks, Series and parallel resonance. Numerical problems.

UNIT-V: Three Phase Circuits:

Advantages of Three Phase Circuits, Balanced and Unbalanced systems, Relation between Line and Phase Quantities in Star and delta connected circuits, Analysis of Balanced & Unbalanced Three Phase Circuits.

Text Books:

1. Engineering Circuit Analysis By W.H. Hayt Jr ,J.E. Kemmerly,S.M.Durbin 8th Ed.,Mc.Graw Hill.
2. Network Analysis: Van Valkenburg; Prentice-Hall of India Private Ltd
3. Basic Electrical Engineering by V.K Mehta & Rohit Mehta- 2006 Edition S.Chand & Company

Reference Books:

1. Circuit Theory Analysis and Synthesis., 7th Edition 2014 by Abhijit Chakrabarthy,Dhanpat Rai&Co.
2. Fundamentals of Electric circuits by Charles k Alexander & M N O Sadiku 5thEd., Mc.Graw Hill.
3. Circuits & Networks by Sudhakar & Syam Mohan, 4th edition, 2010 Mcgraw Hill.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING
(For ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To learn the basic principles of electrical and magnetic circuits and analyze the networks
2. To understand the principle of operation and construction details of DC machines.
3. To understand the principle of operation and construction details of Transformers.
4. To understand the principle of operation and construction details of three phase Induction motor and Alternators.
5. To study the operation of PN junction diode, Half -wave Bridge rectifiers, PNP, NPN Transistors and various Amplifiers.

Course Outcomes:

1. Able to analyze the various Electrical networks and understand the basics of Magnetic Circuits
2. Able to understand the operation of DC generators,3-Point starter and conduct the Swinburne's test.
3. Able to analyze the Performance of Transformers.
4. Able to explain the operation of three phase induction motors and alternator.
5. Able to analyze the operation of Half-wave and Full-wave rectifiers and single stage CE amplifier.

SYLLABUS**UNIT I: Electrical and Magnetic Circuits:**

Basic definitions, Types of network elements, Ohm's Law, Kirchhoff's Laws, Series and parallel Circuits and star-delta and delta-star transformations-simple problems. Magnetic flux, MMF, Reluctance, Faraday's laws, Lenz's law, statically induced EMF, dynamically induced EMF.

UNIT-II: DC Machines:

Principle of operation of DC generator- EMF equation-Types of DC Generators-DC motor Types-Torque equation-Applications-Swinburne's Test, Speed control methods.

UNIT-III: Transformers:

Principle of operation of Single phase Transformers- EMF equation-losses-OC and SC Tests-Efficiency and Regulation.

UNIT-IV: AC Machines:

Principle of operation of Three phase Induction motor-Slip-Torque characteristics-Efficiency-applications- Principle of operation of Alternator-EMF equation, Regulation of alternator by synchronous Impedance method.

UNIT-V: Diodes-Rectifiers and Transistors:

PN junction diode-Forward bias and reverse bias operation, V-I characteristics-Diode applications (Half wave, Full wave and bridge rectifier), Zener diode.

PNP and NPN junction Transistors, Transistor as an amplifier, single stage CE amplifier, Frequency response of CE amplifier.

Text Books:

1. Electrical Technology by Surinder Pal Bali, Pearson Publications. January 1, 2013
2. Electronic Devices and Circuits, R.L Boylestad and Louis Nashelsky, 9th Edition, PEI/PHI 2006.
3. Principles of Electrical Engineering and Electronics, V.K mehta, Published December 1st 2006 by S. Chand & Company Ltd.
4. Principles of Electronics Engineering, V.K mehta, Rohit Mehta, S. Chand Publications. Revised Edition 2012

Reference Books:

1. Basic Electrical Engineering, by J.B Gupta, K Kataria and Sons; Reprint 2013 edition.
2. Fundamentals of Electrical Engineering and Electronics, by Theraja B.L, S Chand Multicolor edition (1 December 2006)

**ENGINEERING PHYSICS LAB
(Common to CIV, EEE & ME)**

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

Training the field oriented Engineering graduates to handle instruments and their design methods to improve the accuracy of measurements.

1. To impart hands-on experience to the students entering engineering/Technology education about handling sophisticated equipment/ instruments.
2. To make the students understand the theoretical aspects of various phenomena experimentally.

Course Outcomes:

Physics lab curriculum gives fundamental understanding of design of an instrument with targeted accuracy for physical measurements.

1. Students get hands on experience in setting up experiments and using the instruments/equipment individually.
2. Get introduced to using new/ advanced technologies and understand their significance.

**LIST OF EXPERIMENTS
(Any 10 of the following listed experiments)**

1. Determination of wavelength of a source-Diffraction Grating-Normal incidence.
2. Newton's rings – Radius of Curvature of Plano - Convex Lens.
3. Determination of thickness of a spacer using wedge film and parallel interference fringes.
4. Determination of Rigidity modulus of a material- Torsional Pendulum.
5. Determination of Acceleration due to Gravity and Radius of Gyration-Compound Pendulum.
6. Melde's experiment – Transverse and Longitudinal modes.
7. Verification of laws of vibrations in stretched strings – Sonometer.
8. Determination of velocity of sound – Volume Resonator.
9. L- C- R Series Resonance Circuit.
10. Study of I/V Characteristics of Semiconductor diode.
11. I/V characteristics of Zener diode.
12. Characteristics of Thermistor – Temperature Coefficients.
13. Magnetic field along the axis of a current carrying coil – Stewart and Gee's apparatus.
14. Energy Band gap of a Semiconductor p - n junction.
15. Hall Effect in semiconductors.
16. Time constant of CR circuit.
17. Determination of wavelength of laser source using diffraction grating.
18. Determination of Young's modulus by method of single cantilever oscillations.
19. Determination of lattice constant – lattice dimensions kit.
20. Determination of Planck's constant using photocell.
21. Determination of surface tension of liquid by capillary rise method.

Reference Book:

1. Advanced Practical Physics –Vol 1 & 2 By S.Venkata Raman, S.Chand & Co.

ENGLISH COMMUNICATION SKILLS LAB- II
(Common to All Branches)

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To enable the students to learn demonstratively the communication skills of listening, speaking, reading and writing.
2. To enable the students participate in group interactions.
3. To improve the presentation skills of the students.
4. To help the students gain their confidence in attending the interviews.

Course Outcomes:

1. A study of the communicative items in the laboratory will help the students become successful in the competitive world.
2. Students enhance their presentation skills.
3. Students participate in group discussions and improve their team skills.
4. Students confidently face the interviews.

SYLLABUS

- ❖ WHY study Spoken English?
- ❖ Making Inquiries on the phone, thanking and responding to Thanks - Practice work.
- ❖ Responding to Requests and asking for Directions - Practice work.
- ❖ Asking for Clarifications, Inviting, Expressing Sympathy, Congratulating
- ❖ Apologising, Advising, Suggesting, Agreeing and Disagreeing - Practice work.
- ❖ Letters and Sounds-Practice work.
- ❖ The Sounds of English-Practice Work
- ❖ Pronunciation
- ❖ Stress and Intonation-Practice work.

Lab Manual:

1. **'INTERACT:** English Lab Manual for Undergraduate Students' Published by Orient Blackswan Pvt Ltd.

Reference Books:

1. Strengthen your communication skills by Dr M Hari Prasad, Dr Salivendra Raju and Dr G Suvarna Lakshmi, Maruti Publications.
2. English for Professionals by Prof Eliah, B.S Publications, Hyderabad.
3. Unlock, Listening and speaking skills 2, Cambridge University Press
4. Spring Board to Success, Orient BlackSwan
5. A Practical Course in effective english speaking skills, PHI
6. Word power made handy, Dr shalini verma, Schand Company
7. Let us hear them speak, Jayashree Mohanraj, Sage texts
8. Professional Communication, Aruna Koneru, Mc Grawhill Education
9. Cornerstone, Developing soft skills, Pearson Education.

C PROGRAMMING LAB
(Common to CIV, EEE & ME)

Lab : 3 Periods

Exam : 3 Hrs.

Int.Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. Understand the basic concept of C Programming, and its different modules that includes conditional and looping expressions, Arrays, Strings, Functions, Pointers, Structures and File programming.
2. Acquire knowledge about the basic concept of writing a program.
3. Role of constants, variables, identifiers, operators, type conversion and other building blocks of C Language.
4. Use of conditional expressions and looping statements to solve problems associated with conditions and repetitions.
5. Role of Functions involving the idea of modularity.

Course Outcomes:

1. Apply and practice logical ability to solve the problems.
2. Understand C programming development environment, compiling, debugging, and linking and executing a program using the development environment.
3. Analyzing the complexity of problems, Modularize the problems into small modules and then convert them into programs.
4. Understand and apply the in-built functions and customized functions for solving the problems.
5. Understand and apply the pointers, memory allocation techniques and use of files for dealing with variety of problems.
6. Document and present the algorithms, flowcharts and programs in form of user manuals.
7. Identification of various computer components, Installation of software

List of Programs

Exercise - 1 Basics

- a) What is an OS Command, Familiarization of Editors - vi, Emacs
- b) Using commands like mkdir, ls, cp, mv, cat, pwd, and man
- c) C Program to Perform Adding, Subtraction, Multiplication and Division of two numbers From Command line

Exercise - 2 Basic Math

- a) Write a C Program to Simulate 3 Laws at Motion
- b) Write a C Program to convert Celsius to Fahrenheit and vice versa

Exercise - 3 Control Flow - I

- a) Write a C Program to Find Whether the Given Year is a Leap Year or not.
- b) Write a C Program to Add Digits & Multiplication of a number

Exercise – 4 Control Flow - II

- a) Write a C Program to Find Whether the Given Number is
 - i) Prime Number
 - ii) Armstrong Number
- b) Write a C program to print Floyd Triangle
- c) Write a C Program to print Pascal Triangle.

Exercise – 5 Functions

- a) Write a C Program demonstrating of parameter passing in Functions and returning values.
- b) Write a C Program illustrating Fibonacci, Factorial with Recursion without Recursion.

Exercise – 6 Control Flow – III)

- a) Write a C Program to make a simple Calculator to Add, Subtract, Multiply or Divide Using switch...case
- b) Write a C Program to convert decimal to binary and hex (using switch call function the function)

Exercise – 7 Functions - Continued

Write a C Program to compute the values of $\sin x$ and $\cos x$ and e^x values using Series expansion. (use factorial function)

Exercise – 8 Arrays

Demonstration of arrays

- a) Search-Linear.
- b) Sorting-Bubble, Selection.
- c) Operations on Matrix.

Exercises - 9 Structures

- a) Write a C Program to Store Information of a Movie Using Structure
- b) Write a C Program to Store Information Using Structures with Dynamically Memory Allocation
- c) Write a C Program to Add Two Complex Numbers by Passing Structure to a Function

Exercise - 10 Arrays and Pointers

- a) Write a C Program to Access Elements of an Array Using Pointer
- b) Write a C Program to find the sum of numbers with arrays and pointers.

Exercise – 11 Dynamic Memory Allocations

- a) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using malloc () function.
- b) Write a C program to find sum of n elements entered by user. To perform this program, allocate memory dynamically using calloc () function.

Understand the difference between the above two programs

Exercise – 12 Strings

- a) Implementation of string manipulation operations **with** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare
- b) Implementation of string manipulation operations **without** library function.
 - i) copy
 - ii) concatenate
 - iii) length
 - iv) compare

Exercise -13 Files

- a) Write a C programming code to open a file and to print its contents onscreen.
- b) Write a C program to copy files

Exercise - 14 Files Continued

- a) Write a C program merges two files and stores their contents in another file.
- b) Write a C program to delete a file.

Note:

- a) All the Programs must be executed in the Linux Environment. (Mandatory)
- b) The Lab record must be a print of the LATEX (.tex) Format.

ENGINEERING PHYSICS - VIRTUAL LABS – ASSIGNMENTS
(Common to CIV, EEE & ME)

Lab : 2 Periods

Int.Marks: 5

Course Objective: Training Engineering students to prepare a technical document and improving their writing skills.

Course Outcome: Physics Virtual laboratory curriculum in the form of assignment ensures an engineering graduate to prepare a /technical/mini-project/ experimental report with scientific temper.

List of Experiments

1. Hall Effect
2. Crystal Structure
3. Hysteresis
4. Brewster's angle
5. Magnetic Levitation / SQUID
6. Numerical Aperture of Optical fiber
7. Photoelectric Effect
8. Simple Harmonic Motion
9. Damped Harmonic Motion
10. LASER – Beam Divergence and Spot size
11. B-H curve
12. Michelson's interferometer
13. Black body radiation

URL: www.vlab.co.in

(Note: Internal Marks of **Engineering Physics - Virtual Labs – Assignments** are to be considered as Assignment marks in the Internal Marks of **Engineering Physics- B17 BS 1204**)

Code: B17 BS 1211

NCC
(Common to CIV, EEE & ME)

Practice : 2 Periods

The NCC- National Integration and Awareness- Drill- Personality Development Life Skills- Leadership- Disaster Management-Social Awareness and Community Development- Health and Hygiene- Environment Awareness and Conservation.

(Note: It is an uncredited course. It will not be included in the Grade Memo / Certificate. The Certificate will be issued based on the performance and attendance. This course attendance will be counted in the semester overall attendance.)



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

II/IV B.TECH

(With effect from **2017-2018** Admitted Batch onwards)

MECHANICAL ENGINEERING

I-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Internal Marks	Exam Marks	Total Marks
B17 BS 2101	Mathematics-IV	3	3	1	--	4	30	70	100
B17 ME 2101	Strength of Materials	3	3	1	--	4	30	70	100
B17 ME 2102	Thermodynamics	3	3	1	--	4	30	70	100
B17 ME 2103	Manufacturing Process	3	3	1	--	4	30	70	100
B17 ME 2104	Metallurgy & Materials Science	3	3	1	--	4	30	70	100
B17 ME 2105	Advanced Engineering Drawing	3	2	--	4	6	30	70	100
B17 ME 2106	Mechanical Engineering Lab	2	--	--	3	3	50	50	100
B17 EE 2107	Basic Electrical & Electronics Engineering Lab	2	--	--	3	3	50	50	100
B17 ME 2107	Auto CAD	1	--	--	2	2	50	--	50
B17 BS 2107	English Proficiency-I	--	1	1	--	2	--	--	--
B17 BS 2108	Professional Ethics & Human Values	--	2	--	--	2	--	--	--
Total		23	20	6	12	38	330	520	850

MATHEMATICS IV
(Common to CE,ECE,EEE& ME)

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students should learn

1. The concept of Analytic function, its implications and applications in flow problems.
2. Solution of one- dimensional wave equation, one-dimensional heat equation and two-dimensional Laplace equation by the use of 'separation of variables'.
3. Formation and solution of linear difference equations. Important concepts of Z-transform and its use to solve linear difference equations.
4. Basic concepts of certain discrete and continuous probability distributions.
5. Concepts of Sampling theory for analyzing large and small samples.

Course Outcomes:

Students will be capable of

1. Using the concept of Analytic function in applications including Electrostatics and Fluid dynamics.
2. Finding theoretical solution of certain Elliptic, Parabolic and Hyperbolic partial differential equations.
3. Using Z-transforms to solve linear difference equations with constant coefficients.
4. Fitting of probability frequency distribution to a given data.
5. Using the concepts of sampling theory to analyze data related to some large and small samples.

SYLLABUS

UNIT-I Functions of a Complex Variable

Review- Cartesian form and polar form of a complex variable, Real and imaginary parts of z^n , e^z , $\sin z$, $\sinh z$ and $\log z$ (**no questions may be set**).

Limit and continuity of a function of the complex variable, derivative, analytic function, entire function, Cauchy- Riemann equations, finding an analytic function, Milne-Thomson method, Applications of analytic function to flow problems, and in Electrostatics. Conformal mapping: the transformations defined by $w = z+c$, $w = cz$, $w = 1/z$, the Bilinear transformation, $w = z^2$ and $w = e^z$.

UNIT-II Applications of Partial Differential Equations

Method of separation of variables, One –dimensional wave equation, the D'Alembert's solution, one-dimensional heat equation, two-dimensional heat flow in steady state (solution of two-dimensional Laplace equation in Cartesian coordinates only)

UNIT-III Difference Equations And Z-Transforms

Formation of a difference equation, Rules for finding complimentary function and particular integral for linear difference equations.

Definition of Z- transform, some standard Z- transforms, properties, transform of a function multiplied by n, initial value theorem and final value theorem(without proof), evaluation of inverse Z- transforms, convolution theorem (without proof), solution of linear difference equations by the use of Z- transforms.

UNIT-IV Probability Distributions

Binomial distribution, Poisson distribution, Normal distribution: Definition (pmf/pdf), notation, mean, variance, moment generating function, probability generating function and fitting of a distribution.

UNIT-V Sampling Theory

Sampling theory: Sampling distribution, standard error, testing of Hypothesis, level of significance, confidence limits, simple sampling of attributes, sampling of variables, estimation of mean and variance.

Large samples: testing of hypothesis for sample proportion, two proportions, single mean and two means.

Small samples: Degrees of freedom, Students' t- distribution, t-test for single mean, two means; Chi-squared distribution-testing the goodness of a fit.

Text Book:

1. Scope and Treatment as in "Higher Engineering Mathematics", by Dr.B.S.Grewal, 43rd Edition, Khanna Publishers.

Reference Books:

1. Advanced Engineering Mathematics, by Erwin Kreyszig, Wiley.
2. A text book of Engineering Mathematics, by N.P.Bali and Dr. Manish Goyal, Lakshmi Publications.
3. Advanced Engineering Mathematics, by H.K.Dass, S.Chand Company.
4. Higher Engineering Mathematics, by B.V.Ramana, Tata Mc Graw Hill Company.
5. Higher Engineering Mathematics, by Dr. M.K.Venkatraman, The National Publishing Company.

STRENGTH OF MATERIALS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Course objective is to learn the fundamental concepts of stress, strain, and deformation of solids with applications to bars, beams, and columns. Fundamentals of applying equilibrium, compatibility, and force-deformation relationships to structural elements are emphasized. This will help students to distinguish different fractures. Different methods are introduced to solve structural analysis problems. Detailed study of engineering properties of materials is also of interest. The course builds on the fundamental concepts of engineering mechanics course. The students will:

1. Gain a fundamental understanding of the concepts of stress and strain by analysis of solids and structures.
2. Study engineering properties of materials and stress-strain relationship.
3. Learn fundamental principles of equilibrium, compatibility, and force-deformation relationship, and principle of superposition in linear solids and structures.
4. Determine stress, strain, deformation of bars, trusses, and beams; stress and strain transformation
5. Analyze determinate and indeterminate axial structural members, torsional members, and beams, torque, shear forces, and bending moments.
6. Be able to perform structural analysis by hand computations and by using computer software.

Course Outcomes:

After successful completion of the course, the student will be able to:

1. Understanding the concepts and determining the stress and strain of simple structures.
2. Locating the Principal Planes and determining the Principal Stresses.
3. Determining Shear Forces and Bending Moments of determinate beams.
4. Determining the distribution of Bending and Shear Stresses of beams.
5. Finding relation between elastic constants. Determining shear stresses due to torsion.
6. Determining stresses in Thin Cylindrical and Spherical shells and Thick Cylinders

SYLLABUS**UNIT-I**

Simple Stresses: Stress, Strain, Stress-Strain curve, Lateral strain, Bars of varying cross-section, Compound bars, Temperature stresses in bars, Modulus of Rigidity, Complementary Shears.

Complex Stresses: Stresses acting on an inclined plane under uniaxial and biaxial state of stress, Principal planes and Principal stresses, Mohr's circle for biaxial stresses.

UNIT-II

Shear Forces and Bending Moments: Beam - Types of loads, Types of supports, types of beams, Shear Force and Bending Moment, S.F. and B.M. diagrams for cantilever, simply supported and over hanging beams loaded with point loads, Uniform Distributed loads and Moments, Relationship between Rate of Loading, Shear Force and Bending Moment.

UNIT-III

Stresses in Beams: Theory of bending, Flexural formula, Determination of bending stresses- section modulus of rectangular, circular, I, and T sections, Determination of simple beam sections, Shear stresses in beams, shear stresses distribution across various beams sections like rectangular, circular, I and T.

UNIT-IV

Elastic Constants and Strain Energy: Bulk modulus, Relationship between elastic constants, Strain energy, Impact Load.

Torsional Stresses in Shafts: Analysis of torsional stresses, Power transmitted, combined bending and torsion

UNIT-V

Thin Cylinders and Spherical Shells: Stresses and strains in thin cylinders, thin spherical shell- derivation for longitudinal and circumferential stresses and volumetric strains.

Thick Cylinders: Lamé's equation- Cylinders subjected to inside and outside pressures- compound cylinders.

Text Books:

1. Analysis of Structures, by Vazirani and Ratwani, Vol. 1, 1993 edition.
2. Solid Mechanics, by Popov

Reference Books:

1. Strength of Materials, by Timoshenko.
2. Strength of Materials -By Jindal, Umesh Publications.
3. Mechanics of Structures Vol-III, by S.B.Junnarkar.

THERMODYNAMICS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives

1. To educate students about the behavior of real gases and the significance of ideal gas theory
2. To educate the students about the properties of ideal gas and their relationship
3. To familiarize the students about various laws of energy transfer occurring and illustrate directional law of nature.
4. To educate the students about the working principle of combustion engines (internal and external) and their cycles such as Otto, Diesel, Atkinson, Ericson, Brayton, etc., and their comparison

Course Outcomes

1. Students realize the practical importance of ideal gas theory and the use of real gases in combustion engines such as IC Engines and Gas turbines
2. Students are able to calculate the properties of the gases such as internal energy, enthalpy and entropy.
3. Students are able to estimate the losses which occur during operation of the heat engines, and their maximum possible operating efficiencies under STP conditions.
4. Students can estimate the maximum work-output delivered by the heat engines and maximum work consumed by the reversed heat engines

SYLLABUS**UNIT – I**

Introduction: Basic Concepts: System, boundary, Surrounding, control volume, Universe, Types of Systems, Macroscopic and Microscopic viewpoints, Concept of Continuum, Thermodynamic Equilibrium, State, Property, Process, Cycle – Reversibility – Quasi – static Process, Irreversible Process, Causes of Irreversibility – Energy in State and in Transition, Types, Work and Heat, Point and Path function. Zeroth law of thermodynamics, Concept of equality of temperatures- Equation of state- Universal gas constant.

UNIT – II

First law of thermodynamics: Joule's experiments-First law of thermodynamics- Isolated systems and steady flow systems- Specific heats at constant volume and pressure - Enthalpy- First law applied to flow systems- Systems undergoing a cycle and change of state- First law applied to steady flow processes-various non-flow processes-Properties of end states- Heat transfer and work transfer- Change in internal energy-throttling and free expansion- Flow processes- Deviations from perfect gas model-Vanderwall's equation of state- Compressibility charts- Variable specific heats

UNIT – III

Second law of thermodynamics-Limitations of the First Law – Thermal Reservoir, Heat Engine, Heat pump, Parameters of performance, Second Law of Thermodynamics, Kelvin-Planck and Clausius Statements and their Equivalence / Corollaries, PMM of Second kind, Carnot's principle, Carnot cycle and its specialties, Clausius theorem Clausius Inequality, Entropy, Principle of Entropy Increase –Third Law of Thermodynamics

UNIT – IV

Air standard Cycles-Otto, Diesel, Dual Combustion cycles, Sterling Cycle, Atkinson Cycle, Ericsson Cycle, brayton cycle – Description and representation on P–V and T-S diagram, Thermal Efficiency, Mean Effective Pressures on Air standard basis – Comparison of Otto-Diesel and Dual cycles based on same compression ratio- same maximum pressure and same maximum temperature.

UNIT – V

General Relations, Availability and Unavailability-Helmholtz function and Gibbs function, Maxwell's equations- Tds relations, relation between specific heats, Available energy, unavailable energy, Available and unavailable forms of energy for a flow and non-flow process-irreversibility

Text Books:

1. Engineering Thermodynamics, by P.K. Nag, Tata McGraw-Hill Publications Company.
2. Thermal Engineering by R.K Rajput, Laxmi publications.
3. Applied Thermodynamics-I by R. Yadav, Central Book House.

References Books:

1. Engineering Thermodynamics by Rathakrishnan, Prentice - Hall India.
2. Engineering Thermodynamics by Y.V.C. Rao.
3. Engineering Thermodynamics by K. Ramakrishna, Anuradha agencies.
4. Engineering Thermodynamics Work and Heat Transfer, by G.F.C Rogers and Y.R. Mayhew, ELBS publication.

MANUFACTURING PROCESS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To provide an understanding and appreciation of the different manufacturing and fabrication methods to the students.
2. To expose the students to various applications of the manufacturing process in the real life articles/products.

Course Outcomes:

1. Student will be able to recognize various manufacturing materials, manufacturing process and types of productions.
2. Student will be able to identify various casting processes, metal forming process and welding process.
3. Student will be able to design of gating system, patterns and cores for various casting processes.
4. Student will be able to apply knowledge of casting process for manufacturing of products.
5. Student will be able to apply knowledge of rolling, forging, extrusion for manufacturing of products.
6. Student will be able to apply knowledge of welding, brazing and soldering for joining of metals.

SYLLABUS**UNIT – I**

Manufacturing concepts: Product cycle, Job, batch and mass production, Primary and secondary manufacturing processes.

Metal Casting Process: Principle of metal casting, Pattern: Materials, Allowances and Types, Core boxes, Moulding sands: ingredients, properties, preparation, types, Moulding tools, Sand testing, Sand moulding, Machine moulding, Core making, Melting and pouring- Classification of furnaces, Cupola furnace, pouring laddels; Element of gating system, casting defects.

UNIT – II

Special Casting Techniques: Permanent mould casting, Pressure die casting, Centrifugal casting, Shell mold casting, Investment casting and CO₂ process.

UNIT – III

Metal Forming: Hot & Cold working, Rolling, Extrusion, metal spinning, Drawing, Piercing.

Sheet Metal Forming: Concept of spring back, Materials, tools, operations, embossing, coining, stretch forming, Progressive and Compound Dies.

UNIT – IV

Forging Processes: Forgability, Forging Materials, Classification: smith, drop, press and machine forging, Forging tools, Forging Operations, High energy rate forming, Swaging.

UNIT – V

Welding Processes: Welding metallurgy, Weldability, Classification: Plastic welding (Forge, Resistance &Thermit welding), Fusion welding (Gas, Arc & Thermit welding), Solid state welding (Friction, Ultrasonic, Diffusion and Explosive welding), Soldering and Brazing, Weld defects, Weld inspection and testing.

Text Books:

1. Elements of Workshop Technology Vol-1: Manufacturing Processess by S.K. Hajra Choudhury, A.K. Hajra Choudhury,Nirjhar Roy, MPP, Pvt. Ltd.
2. Manufacturing Technology- Foundary, Forming and Welding by P.N. Rao, Tata McGraw- Hill Publishing Company.

Reference Books:

1. Process and Materials of Manufacture (4th Edition) by Roy A. Lindberg, Prentice-Hall of India Private Limited.
2. Manufacturing Engineering & Technology by Kalpak Jain, Addition Wesley Edition.
3. Materials and Processes in Manufacturing by De Margo, Black and Kohsen, Prentice Hall of India.
4. Principles of Metal Casting by Hein and Rosenthol, Tata Mc-Graw Hill India.

METALLURGY & MATERIALS SCIENCE

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The overall objective of the course is to impart knowledge about the engineering materials and their properties and predict their behavior under different working conditions and methods to enhance the mechanical properties of the material Specific objectives include:

1. This course will help in learning crystal structures, crystal defects, space lattices.
2. This course will help students to understand systems of solid solutions and their nature.
3. To acquaint the knowledge about the cooling curves and phase diagrams of different alloy systems, calculation of weight percent of phases in alloy system by applying Lever rule.
4. To acquaint knowledge about ferrous alloys, particularly Steel and Cast Irons.
5. To impart knowledge about Equilibrium and Non-Equilibrium diagrams.
6. To impart knowledge about different heat treatment and surface hardening methods in improving the mechanical properties of steels.
7. To impart knowledge about the plastic deformation of materials.
8. To impart knowledge about composite materials.

Course Outcomes:

Upon successful completion of this course student should be able to:

1. Understand crystalline solids and their atomic structures.
2. Suggest and recommend necessary engineering materials for specific applications keeping in view of the cost, design, reliability, life, working conditions and properties of the products.
3. Understand different phase transformations in Iron-Iron Carbide diagram and distinguish between steels and cast irons.
4. Select different materials for tools and components based on functional requirements.
5. Use composite materials for different engineering applications like aerospace, automobile, ship building industry, sports item etc.
6. Inclination towards self learning, higher education and research work in the field of engineering materials.

SYLLABUS**UNIT-I**

Structure of crystalline solids: Atomic structure & bonding in solids- Unit cell, Space lattice, Crystal structures and its types-calculations of radius, Coordination Number and Atomic Packing Factor for different cubic structures - Imperfection in solids, point defects, Line defects, Planar defects and Volume defects- Concept of Slip & twinning, Indices for planes and directions.

UNIT-II

Phase diagrams: Basic terms-Solid solutions - Gibbs phase rule- Lever rule – cooling curves- Phase diagrams - construction of phase diagrams- binary phase diagrams - Al-Cu and AlSi phase diagrams- Invariant reactions, eutectic, peritectic, eutectoid, peritectoid, metatectic & monotectic reactions, Iron carbon phase diagram -Heat treatment of steel- Annealing, and its types, normalizing, hardening, tempering, martempering, austempering

UNIT-III

TTT diagrams, Construction of TTT diagram, TTT diagram for hypoeutectoid and alloy steels, CCT diagram- Martensitic transformation, nature of martensitic transformation- Surface hardening processes like case hardening, carburizing, cyaniding, nitriding Induction hardening, Flame hardening, hardenability, Jominy end-quench test, Precipitation Hardening

UNIT-IV

Engineering Alloys: Effect of alloying elements of steel -Properties, composition, and uses of Plain carbon, low carbon, medium & high carbon steels. stainless steels, high speed steels, Hadfield steels, tool steels - Cast irons, gray CI, white CI, malleable CI, SC iron-The light alloys- Al & Mg & Titanium alloys- Copper & its alloys: brasses & bronzes- super alloys, Smart materials- Nano materials.

UNIT-V

Composite Materials: Classification of composite materials, dispersion strengthened, particlereinforced and fiber reinforced composite laminates properties of matrix and reinforcement materials and structural applications of different types of composite materials – Types of Fabrication of composite materials. Powder Metallurgy: Production of metal powders - Powder Metallurgy process and its applications

Text Books:

1. “Materials Science & Engineering- An Introduction”, William D.Callister Jr. Wiley India Pvt. Ltd. 6th Edition, 2006, New Delhi.
2. “Material Science and Metallurgy for Engineers”, Dr. V.D Kodgire and S.V Kodgire
3. Physical Metallurgy, Principles & Practices”, V Raghavan.PHI 2nd Edition 2006, New Delhi.

Reference Books:

1. Introduction to Physical Metallurgy by Sidney H AvnerTata McGraw-Hill Education 1997.
2. Materials Science And Engineering: A First Course By V. Raghavan Phi 5th Edition 2011, New Delhi.

ADVANCED ENGINEERING DRAWING

Lecture	: 2 Periods	Int. Marks	: 30
Tutorial	: 4 Period	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives

1. To highlight the significance of universal language of engineers.
2. To impart basic knowledge and skills required to prepare engineering drawings.
3. To impart knowledge and skills required to draw projections of solids in different contexts.
4. To visualize and represent the pictorial views with proper dimensioning and scaling.

Course Outcomes

Students will be able to:

1. Apply principles of drawing to represent dimensions of an object.
2. Draw projections solids with axis inclined to both planes.
3. Represent sectional views of solids.
4. Develop the surfaces of regular solids and draw the projections of intersection of solids.
5. Gain knowledge on Computer Aided Drafting.

SYLLABUS**UNIT-I**

Projection of solids: projection of solids with axis inclined to both the reference planes.

UNIT-II

Projections of Section of Solids: Section Planes: Parallel and inclined section planes, Sections and True shape of section, Sections of Solids: Prism, Pyramid, Cylinder and Cone.

UNIT-III

Development of Surfaces: Methods of Development: Parallel line development and radial line development. Development of a cube, prism, cylinder, pyramid and cone.

UNIT-IV

Interpenetration of Right Regular Solids: Intersection of Cylinder Vs Cylinder, Prism Vs Prism, Prism Vs Cylinder, Cylinder Vs Cone.

UNIT-V

Computer Aided Drafting: Introduction on Computer Aided Drafting, Display Devices, Input devices, Output devices, Introduction on Auto CAD, Different Commands used in Auto CAD, Brief discussion on Geometric Modelling.

Text Book:

1. Elements of Engineering Drawing by N.D. Bhatt (52nd Edition, 2013).

Reference Book:

1. Engineering Graphics by K.L. Narayana and P. Kannaiah

MECHANICAL ENGINEERING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To understand the principle and functioning of various mechanical devices such as boilers, engines etc.
2. Ability to understand the working of two stroke and four stroke engines.
3. Acquiring the knowledge of operation of a reciprocating compressor and to assess the pressure gauge performance.
4. The way of determination of flash and fire points of oil samples and carbon residue and their importance is acquired.
5. The procedure for determination of calorific values of the fuels and viscosities of oil samples can be understood.
6. Practically the procedure for moment of inertia of fly wheel, connecting rod and modulus of rigidity is acquired.

Course Outcomes:

1. Students are now aware of the use of drawing valve timing diagrams of an engine and method to evaluate the volumetric efficiency of air compressor.
2. They are also aware of method of calibrating pressure gauge, the importance of flash and fire points and calorific values of fuels.
3. The importance and application by calculating viscosities of oil samples are understood.
4. The use of moment of inertia and modulus of rigidity is understood.
5. They are also now able to identify the parts of boiler and engines etc.

LIST OF EXPERIMENTS

1. Study and valve timing diagrams for four-stroke and study & PTD of two-stroke engines.
2. Determination of volumetric efficiency of the given air compressor by (i) plate orifice method and (ii) tank capacity method.
3. Calibration of the given pressure gauge.
4. Determination of flash and fire points and b) Canradsons carbon residue test.
5. Determination of calorific value of flues (solid, liquid and gaseous) by Bomb calorimeter/Gas calorimeter.
6. Determination of the kinematic and absolute viscosity of the given sample oils.
7. Determination of inertia of the given flywheel and connecting rod.
8. Determination of modulus of rigidity of the given wire with torsion pendulum.
9. Study of boilers, various mountings and accessories.
10. Assembling of the given two-stroke petrol engine. (Instead of engine, any mechanical unit can be given for this experiment.)

Reference Books:

1. Engineering Mechanics by S.Timoshenko and D.HYoung McGraw-Hill.
2. Engineering Mechanics by Singer.
3. Internal Combustion Engines by V. Ganesan, McGraw-Hill.

BASIC ELECTRICAL AND ELECTRONICS ENGINEERING LAB

Lab	: 3 Periods	Int. Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 2

Course Objectives:

1. Information to supplement to the Electrical & Electronics Engineering courses.
2. The ability to conduct testing and experimental procedures on Circuits.

Course Outcomes:

Students will be able to

1. Apply the concepts of Theorems for a given electrical circuit.
2. Evaluate the efficiency and regulation of a single phase transformer.
3. Relate physical observations and measurements involving electrical circuits' theoretical principles.
4. Design amplifier circuit using NPN transistor

LIST OF EXPERIMENTS**Part-A: Electrical Engineering**

1. Verification of KCL and KVL
2. Verification of Ohms law (draw the V-I characteristics for a particular resistor)
3. Swinburne's test on D.C shunt Machine (predetermination of efficiency when working as motor and generator)
4. Brake test on D.C shunt motor. (determination of performance characteristics)
5. Brake test on D.C series motor. (determination of performance characteristics)
6. Brake test on three phase induction motor.(determination of performance characteristics)
7. Speed control of D.C shunt by (a) Armature voltage control (b) Field flux control Method.
8. OC and SC test on single phase Transformer (predetermination of efficiency and regulation at given power factor).

Part- B: Electronics Engineering

1. PN junction Diode Characteristics (a) Forward bias (b) Reverse bias. (cut in voltage and resistance calculations)
2. Half wave rectifier with and without filters.
3. Full wave rectifier with and without filters.
4. Transistor CE characteristics (Input and Output)
5. Characteristics CE amplifier
6. Zener diode characteristics
7. Regulation characteristics of Zener diode.

Reference Books:

1. Electrical Technology by Surinder Pal Bali, Pearson Publications.
2. Electronic Devices And Circuits, R.L Boylestad and Louis Nashelsky, 9th edition, PEI/PHI 2006.

AUTOCAD

Lab	: 2 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Credits	: 1

Course Objectives:

1. Increase ability to communicate with people
2. Learn to sketch and take field dimensions.
3. Learn to take data and transform it into graphic drawings.
4. Learn basic Auto Cad skills.
5. Learn basic engineering drawing formats
6. Prepare the student for future Engineering positions

Course Outcomes

Up on completion of the course the student shall be able to gain knowledge on:

1. Auto CAD screen and various Tool bars and menus and Explain about Dimensioning and Hatching
2. Draw the 2D – drawings like knuckle joint, screw jack, flange coupling, lathe tool post, eccentric etc.,
3. Explain about 3D solids and solids tool bar options and Drawing of 3D – components like bolt & nut, screw jack
4. Rendering of 3D images

LIST OF EXERCISES

1. Study the Auto CAD screen, various toolbars and menus
2. Exercises on usage of Draw and modify tool bar.
3. Exercises on mirror, rotate, array and move commands
4. Exercises on dimension and hatching
5. Draw the 2D knuckle joint with full details & dimensioning
6. Draw the screw jack 2D drawing
7. Study the 3D solids (primitives) and solids tool bar options
8. Draw bolt and nut in 3D
9. Draw various parts of screw jack in assemble them as 3D component
10. Render the 3D images already generated and apply materials and light.

Reference Books:

1. **Lab Manual.**

ENGLISH PROFICIENCY-I
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

Enriching the communicative competency of the students by adopting the activity-based as well as the class-oriented instruction with a view to facilitate and enable them to enhance their language proficiency skills.

Course Objectives:

The Students will be able to

1. Communicate their ideas and views effectively.
2. Practice language skills and improve their language competency.
3. Know and perform well in real life contexts.
4. Identify and examine their self attributes which require improvement and motivation.
5. Build confidence and overcome their inhibitions, stage freight, nervousness, etc.
6. Improve their innovative practices in speaking.

Course Outcomes:

The Students will

1. Improve speaking skills.
2. Enhance their listening capabilities.
3. Learn and practice the skills of composition writing.
4. Enhance their reading and understanding of different texts.
5. Improve their inter-personal communication skills.
6. Be confident in presentation skills.

SYLLABUS

UNIT-I :LISTENING

Selected Motivational Speeches

Selected Moral Stories

UNIT-II :SPEAKING

Book Review

Skit Presentation

PowerPoint Presentations

Describing event/place/thing

Extempore

Group Discussion

Picture Perception and Describing Test

UNIT-III : READING

Speeded Reading

Reading Comprehension

UNIT-IV :WRITING

Paragraph Writing

Literary Appreciation – Understanding the Language of Literature

UNIT-V :PROJECT

Ad Making

Reference Books:

1. Classic Short Stories-A Reader's Digest Selection
2. English for Colleges by Brendan J. Carroll, Macmillan Publications
3. The World's Great Speeches- edited by B.S. Sekhar, Jeet Publications
4. Fundamentals of Technical Communication by Meenakshiraman, Sangeta Sharma of OUP
5. English and Communication Skills for Students of Science and Engineering, by S.P. Dhanavel, Orient Blackswan Ltd. 2009
6. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
7. The Oxford Guide to Writing and Speaking by John Seely OUP

PROFESSIONAL ETHICS & HUMAN VALUES
(Common to CIVIL,EEE & MECH)

Lecture	: 2 Periods.	Int.Marks	: --
Exam	: ---	Credits	: --

Course Objectives:

1. To inculcate Ethics and Human Values into the young minds.
2. To develop moral responsibility and mould them as best professionals.
3. To create ethical vision and achieve harmony in life.

Course outcomes:

By the end of the course student should be able to understand the importance of ethics and values in life and society.

SYLLABUS

UNIT – I

Ethics and Human Values: Ethics and Values, Ethical Vision, Ethical Decisions, **Human Values** – Classification of Values, Universality of Values.

UNIT – II

Engineering Ethics: Nature of Engineering Ethics, Profession and Professionalism, Professional Ethics
Code of Ethics, Sample Codes – IEEE, ASCE, ASME and CSI.

UNIT – III**Engineering as Social Experimentation:**

Engineering as social experimentation, Engineering Professionals – life skills, Engineers as Managers, Consultants and Leaders Role of engineers in promoting ethical climate, balanced outlook on law.

UNIT – IV**Safety Social Responsibility and Rights:**

Safety and Risk, moral responsibility of engineers for safety, case studies – Bhopal gas tragedy, Chernobyl disaster, Fukushima Nuclear disaster, Professional rights, Gender discrimination, Sexual harassment at work place.

UNIT – V**Global Issues:**

Globalization and MNCs, Environmental Ethics, Computer Ethics, Cyber Crimes, Ethical living, concept of Harmony in life.

Text Books:

1. Govindharajan, M., Natarajan, S. and Senthil Kumar, V.S., Engineering Ethics, Prentice Hall of India, (PHI) Delhi, 2004.
2. Subramainam, R., Professional Ethics, Oxford University Press, New Delhi, 2013.

References:

1. Charles D, Fleddermann, "Engineering Ethics", Pearson / PHI, New Jersey 2004 (Indian Reprint).

SCHEME OF INSTRUCTION & EXAMINATION
(Regulation R17)

II/IV B.TECH
(With effect from **2017-2018** Admitted Batch onwards)

MECHANICAL ENGINEERING

II-SEMESTER

Code No.	Course	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Total Contact Hrs/Week	Sessional Marks	Exam Marks	Total Marks
B17 ME 2201	Advanced Strength of Materials	3	3	1	--	4	30	70	100
B17 ME 2202	Thermal Engineering	3	3	1	--	4	30	70	100
B17 ME 2203	Metal Cutting & Machine Tools	3	3	1	--	4	30	70	100
B17 ME 2204	Fluid Mechanics	3	3	1	--	4	30	70	100
B17 ME 2205	Mechanical Engineering Drawing	3	--	--	4	4	30	70	100
B17 BS 2203	Engineering Economics	3	3	1	--	4	30	70	100
B17 ME 2208	Manufacturing Process Lab	2	--	--	3	3	50	50	100
B17 CE 2210	Strength of Materials Lab	2	--	--	3	3	50	50	100
B17 ME 2209	Industry Oriented Technology Lab	1	--	--	2	2	50	--	50
B17 BS 2206	English Proficiency-II	--	1	1	--	2	--	--	--
Total		23	16	6	12	34	330	520	850

ADVANCED STRENGTH OF MATERIALS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To impart students the knowledge about the calculation of slope and deflection of statically determinate beams.
2. To enrich the student on the concepts of shear force and bending moment diagrams of fixed beams with uniform and non-uniform cross sections, both under stability of supports and sinking of supports.
3. To make the student understand the concepts of shear force and bending moment diagrams of continuous beams with uniform and non-uniform cross sections, both under stability of supports as well as sinking of supports.
4. To analyze the stresses produced in practical applications of curved bars.
5. To enhance the knowledge of students on different theories applied for analysis of columns and struts.

Course Outcomes:

Students are able to:

1. Find the slope deflection produced in cantilever, simply supported and overhanging beams subjected to different kinds of lateral loads.
2. Draw the bending moment and shear force diagrams of fixed beams of uniform and non uniform cross sections subjected to different load conditions, and having sinking of support.
3. Draw the bending moment and shear force diagrams of continuous beams subjected to different load conditions, and having sinking of support.
4. Evaluate the stresses across the cross-sections of the curved beam and crane hook subjected to external loads.
5. Apply different theories to analyze the crippling stresses induced in columns and struts subjected to different load conditions.

SYLLABUS**UNIT –I**

Deflections of Beams: Relation between curvature, slope and deflection; Slope and deflection of cantilever, simply supported and overhanging beams – Macaulay's method and Moment area method.

UNIT –II

Fixed Beams: Relations between fixing moments of a fixed beam of uniform cross section, BMD & SFD of fixed beams of uniform and variable cross sections, Effect of sinking of support.

UNIT –III

Continuous beams: Clapeyron's theorem of three moments for a continuous beam of varying and uniform cross sections, BMD & SFD of continuous beams of uniform cross section, Effect of sinking of support.

UNIT –IV

Columns and Struts: Buckling of columns, Euler's theory – Columns with both ends hinged, both ends fixed, one end fixed and other end hinged, one end free and other end fixed, Limitation of Euler's formula, Empirical formulae – Rankine's formula, Column carrying eccentric load.

UNIT-V

Bending of Curved Bars: Winkler-Bach theory of curved bars subjected to uniform bending moment – rectangular, circular, and trapezoidal cross sections, Stresses in a crane hook.

Text Books:

1. Analysis of Structures, Vol. – I by Vazirani and Ratwani, Khanna Publishers
2. Strength of materials by Sadhu Singh, Khanna Publishers

Reference Book:

1. Strength of Materials, by Timoshenko, CBS Publishers and distributors.

THERMAL ENGINEERING

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To gear the student with basic principles of steam properties.
2. To prepare the student for industrial application of steam.
3. The student is taught to design the steam equipment so that R&D in industry is improved.

Course Outcomes:

1. The student gets complete knowledge of steam and its properties.
2. The student learns the complete calculation procedures for designing steam turbines, steam condensers, nozzles etc. used in thermal power plants, steam engines, water turbines and many other industrial applications.
3. The student is prepared to work in industry immediately after his course

SYLLABUS**UNIT-I****Properties of Pure Substance:**

Definition of pure substance, phase change of a pure substance, p-T (Pressure-Temperature) diagram for a pure substance, p-V-T(Pressure-Volume-Temperature) surface, phase change terminology and definitions, property Diagrams in common use, Formation of steam, Important terms relating to steam formation, Thermodynamic properties of steam and steam tables, External work done during evaporation, Internal latent heat, Internal energy of steam, Entropy of water, Entropy of evaporation, Entropy of wet steam, Entropy of superheated steam, Enthalpy-Entropy (h-s) charts for Mollier diagram, Determination of dryness fraction- Tank or bucket calorimeter, throttling calorimeter, separating and throttling calorimeter.

UNIT-II**Vapor Power Cycles**

Vapor power cycle- Rankine cycle- Reheat cycle- Regenerative cycle- Thermodynamic variables effecting efficiency and output of Rankine and Regenerative cycles- Improvements of efficiency, Binary vapor power cycle.

UNIT-III**Steam Nozzles:**

Type of nozzles- Flow through nozzles- Condition for maximum discharge- Nozzle efficiency- Super saturated flow in nozzles- Relationship between area velocity and pressure in nozzle flow- Steam injectors.

UNIT-IV

Steam Turbines:

Classification of steam turbines- Impulse turbine and reaction turbine- Compounding in turbines- Velocity diagrams in impulse and reaction turbines- Degree of reaction- Condition for maximum efficiency of reaction turbines- Effect of friction on turbines constructional features governing of turbines.

UNIT-V

Condensers:

Classification of condenser- Jet, Evaporative and surface condensers- Vacuum and its measurement- Vacuum efficiency- Sources of air leakage in condensers- Condenser efficiency- Daltons law of partial pressures- Determination of mass of cooling water- Air pumps.

Steam Boilers –

Working principle of various boilers their accessories and mountings (Simple vertical, Cochran, Babcock & Wilcox and Lancashire Boiler), Performance of boilers (simple problems)

Text Books:

1. A Treatise on Heat Engineering by Vasandhani and Kumar.
2. Applied Thermodynamics-II by R. Yadav.
3. Thermal Engineering, by R. K. Rajput.

References Books:

1. Fundamentals of Engineering Thermodynamics by E. Radhakrishna, PHI.
2. Fluid Flow Machines, by M.S. Govinda Rao, Tata McGraw Hill publishing company Ltd.
3. Refrigeration and Air-conditioning, by C.P.Arora and Domokundwar.
4. Thermal Science and Engineering by D.S. Kumar, S.K. Kataria and Sons.
5. Refrigeration and Air-conditioning, by Ahamadul Ameen, PHI.

METAL CUTTING & MACHINE TOOLS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To give a clear understanding of the mechanism of machining to the students.
2. To describe the mechanisms of the various machine tools, types of machine tools, various operations that can be performed on them, machining time and force calculations etc to the students.

Course Outcomes:

1. Students will be able to describe the mechanisms of metal cutting.
2. Students will be able to calculate cutting forces, tool life and machining parameters.
3. Students will be able to design the single point and multi point cutting tools.
4. Students will be able to demonstrate the working of various machine tools like lathe, milling machine and grinding machine etc.
5. Students will be able to identify different micro finishing operations.
6. Students will be able to assess the advantages, limitations and applications of unconventional methods of machining.

SYLLABUS**UNIT-I**

Mechanics of Metal Cutting: Orthogonal and oblique cutting, mechanics of chip formation, types of chips; classification, nomenclature, signature (ASA & ISO systems) of single point cutting tools, tool materials; tool wear and tool life; Cutting forces-Merchant's circle, Machinability, Cutting fluids.

UNIT-II

Machine tools using Single point cutting tools: Engine lathe; Capstan and turret lathe, shaper, planner, Slotter and boring-Types, Parts, Specifications, Mechanisms, Operations and machining parameters.

UNIT-III

Machine tools using Multi point cutting tools: Drilling machine-Types, Parts, Specifications, Mechanisms, Types of drills, Nomenclatures of twist drill, Operations and machining parameters

Milling machine-Types, Parts, Specifications, Mechanisms, Attachments, Types of Milling cutters, Nomenclature of plain milling cutter, Operations, machining parameters and Indexing methods.

Broaching machine-Types, Parts, Specifications, Types of Broaches, Nomenclature of pull broach, Operations and machining parameters

UNIT-IV

Machine tools using Abrasive wheels: Grinding Machine- Types, Parts, Specifications, Manufacturing of grinding wheel-bonding processes, grit, grade and structure, selection of grinding wheels, mounting of grinding wheels, glazing, loading, dressing and truing of grinding wheel, Operations and machining parameters

Micro finishing Operations-Lapping, honing, super finishing, polishing and buffing

UNIT-V

Unconventional Methods of Machining: Process, Characteristics, Advantages, Limitations, Applications of Abrasive Jet Machining (AJM), Ultrasonic Machining (USM), Water Jet Machining (WJM), Electro Discharge Machining (EDM), Wire-cut EDM, Electron Beam Machining (EBM), Plasma Arc Machining (PAM), Laser Beam Machining (LBM), Chemical milling; Photochemical milling, Electro Chemical Machining (ECM), Electro Chemical Grinding (ECG)

Text Books:

1. Elements of Workshop Technology Vol-2: Machine Tools by S.K. Hajra Choudhury, A.K. Hajra Choudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2. Metal cutting and Machine tools by P.N. Rao, Tata McGraw- Hill Publishing Company.
3. Process and Materials of Manufacture (4th Edition) by Roy A. Lindberg, Prentice-Hall of India Private Limited.

Reference Books:

1. Fundamentals of Metal Machining and Machine Tools by Geoffrey Boothroyd, International Student Edition, McGraw-Hill Book Company.
2. Metal Cutting Principles by M.C. Shaw, MIT Press, Cambridge.
3. Advanced Methods of Machining by J. A. McGeough, Chapman & Hall Publishers.
4. Metal Cutting-Theory and Practice by Amitabha Bhattacharya, Central Book Publishers.
5. Production Engineering by P.C. Sharma, S. Chand and Company.

FLUID MECHANICS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives

1. develop an appreciation for the properties of Newtonian fluids,
2. study analytical solutions to variety of simplified problems,
3. understand the dynamics of fluid flows and the governing non-dimensional parameters,
4. apply concepts of mass, momentum and energy conservation to flows.

Course Outcomes

After the completion of the course, students are able to

1. Apply the Bernoulli equation to solve problems in fluid mechanics.
2. Apply the concepts of momentum equation for finding the forces acting on the vanes of the turbines.
3. Apply control volume analysis to problems in fluid mechanics.
4. Apply potential flow theory to solve problems in fluid mechanics.
5. Identify the recent developments in fluid mechanics, with application to aerospace systems.

SYLLABUS**UNIT-I**

Properties of fluids- Introduction-Viscosity- Pressure and its measurement, Absolute, Gauge, Atmospheric and Vacuum pressure – Manometers, Simple manometers, Differential manometers. Hydrostatic forces on surfaces- Total Pressure and Pressure Centre- Vertical, Horizontal, inclined and Curved plane surfaces submerged in liquid- Buoyancy and Flotation.

UNIT-II

Fluid Kinematics & Fluid Dynamics: Types of fluid flow- Continuity equation- Velocity potential function and Stream Function- Types of Motion, Linear Translation, Linear deformation, Angular deformation, Rotation, Vorticity and circulation-Vortex flow, forced and Free Vortex – Equation of Motion- Euler's equation - Bernoulli's equation and its applications- Venturimeter, Orifice Meter, Pitot tube-Momentum Equation-Momentum of momentum Equation.

UNIT-III

Viscous Flow: Favourable pressure gradient and adverse pressure gradient-Power absorbed in Viscous Flow- Flow through pipes & Plates- Hagen Poiseuille flow- Darcy's Weisbach friction factor- Loss of head due to friction in pipes, Minor Losses and Major losses - Flow through branched pipes- Power transmission through pipes. **Dimensional and Modeling Analysis:** Fundamental and derived dimensions- Dimensionless groups- Rayleigh method- Buckingham π -theorem- Model Analysis - Types of similarity- Geometric, Kinematic and Dynamic similarities- Dimensionless numbers- Model Laws.

UNIT-IV

Laminar Boundary Layer: Definition- Laminar Boundary Layer- **Turbulent Boundary Layer** -Laminar Sub layer- Boundary Layer thickness-Displacement thickness, Momentum thickness and Energy thickness-Momentum integral equation- Flow over a flat plate.

Turbulent Boundary Layer: Laminar- Turbulent transition- Momentum equations and Renold's stresses- Fully developed turbulent flow through a pipe- Turbulent boundary layer on a flat plate- Laminar sub-layer- Boundary layer separation and control.

UNIT-V

Compressible Fluid Flow: Thermodynamic relations- Continuity, Momentum and Energy equations- Velocity of sound in a compressible fluid- Mach number and its significance- Limits of incompressibility- Pressure field due to a moving source of disturbance- Propagation of pressure waves in a compressible fluids- Stagnation properties- Stagnation pressure, Temperature and density- Area velocity relationship for compressible flow

Text Books:

1. Fluid Mechanics and Hydraulic Machines, by R. K. Bansal, Laxmi publications.
2. Fluid Mechanics, by A.K. Mohanty, Prentice Hall of India Pvt.Ltd.

References:

1. Fluid Mechanics and Fluid Power Engineering by Dr. D.S. Kumar, S.K. Kataria & Sons.
2. Foundations of Fluid Mechanics, by Yuan, Prentice Hall of India.
3. Fluid Mechanics and its Applications, by S. K.Gupta and A.K.Gupta, Tata McGraw Hill, New Delhi.
4. Fluid Mechanics and Hydraulic Machines by R. K. Rajput, S.Chand & Co.
5. Fluid Mechanics by Kothandaraman and Rudramoorthy.

MECHANICAL ENGINEERING DRAWING

Practice : 4 Periods
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives

1. To provide an understanding and draw the assembly and production drawing of various engine components and machine tool components to the students.
2. To expose the students to draw various screw fastenings (Screw, Riveted, welded etc.), Bearings, couplings, key, dimensional and geometrical tolerances etc.

Course Outcomes

Students will be able to

1. Know drawing of Screw threads and Screw Fastenings using standard Empirical formulae.
2. Draw Riveted joints, Keys, Cotter-joint, Draw Couplings (Shaft couplings: Box and split muff couplings, Flanged, Flexible, Universal and Oldham couplings).
3. Draw the dimensional and geometrical tolerances and surface roughness symbols.
4. Draw Assembly and production drawings of various engine components and machine tool components.

SYLLABUS

Screw threads, Screw Fastenings, keys, and Riveted joints using standard Empirical formulae.

Cotter-joints, Shaft couplings: Box and split muff couplings, Flanged, Flexible, Universal and Oldham couplings,

Assembly drawing of various engine components and machine tool components (Simple eccentric, swivel bearing, plumber block, Screw Jack, Stuffing Box).

Conventional representations, Limits, Fits and Tolerances, Geometrical Tolerances, Indication of surface roughness, Production Drawings.

Text Books:

1. Machine Drawing, by N.D.Bhatt, Charotal Publishing House.g
2. Production Drawing by K.L Narayan, P. Kannaiah and K. Venkata Reddy, New Age.
3. Engineering Drawing, by A.C.Parkinson, Wheeler Publishing.

Reference Book:

1. Machine Drawing by K.L Narayan, P. Kannaiah and K. Venkata Reddy, New Age.

ENGINEERING ECONOMICS

Lecture	: 3 Periods	Int.Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objective:

To create awareness on application of economic & accounting concepts in the organization for engineering students. In this regard, they will go through demand analysis, cost concepts, pricing practices, macro- economic concepts & fundamentals of accounting.

Course Outcomes:

1. Provide detailed insight about origin & definitions of economics & enlighten the students about demand analysis.
2. Illustration about applications of cost Concepts & analysis of breakeven point.
3. Understand about various types of Market Structure and Pricing practices implemented by the organization.
4. Infuse knowledge about different Economic systems & Business cycles.
5. Enlighten the students regarding the aspects of Depreciation & Financial Accounting.

SYLLABUS

UNIT-I

Introduction to Economics: Wealth, Welfare and Scarce Definitions of Economics; Micro & Macro Economics.

Demand Analysis: Demand Determinants, Law of Demand and its exceptions. Elasticity of demand – Meaning, types, Significance of Elasticity of Demand, Measurement of price Elasticity of Demand. Need for Demand forecasting, forecasting techniques.

UNIT-II

Cost Analysis: Classification of cost, Elements of cost, Methods of costing (Job costing, Process costing & Unit costing).

Break-Even Analysis(BEA): Determination of Break-Even Point, Assumptions and Applications.

UNIT-III

Market Structures: Features and price determination under Perfect competition, Monopoly, Monopolistic competition and Oligopoly.

Pricing practices: Price - meaning, methods of pricing.

UNIT-IV

Economic Systems: Features and Evaluation of Capitalism, Socialism and Mixed Economy.

Business cycles: Meaning, Phases , Causes & theories of Business Cycle.

UNIT-V

Depreciation and Financial Accounting: Depreciation-causes and methods (straight line method, diminishing balance method).

Final Accounts: Preparation of Trading Account, Profit & Loss Account and Balance sheet.

Text Books:

1. Managerial Economics & Financial Analysis-by Dr.A.R.AryaSri,TMH 2011.
2. Engineering Economics-by Tarachand,Nem Chand &Bros.Roorke.

Reference Books:

1. Modern Economics - by K. K. Dewett,S.Chand&Co,New Delhi.
2. Principles of Economics-Vrinda Publications(P)Ltd.New Delhi.

MANUFACTURING PROCESS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Objectives:

1. To recognize the tools, materials, machines used for making products in foundry, welding and machine shop.
2. To differentiate the different welding techniques used for different materials.
3. To realize the various moulding sands used for making moulds

Course Outcomes:

1. Student will be able to prepare moulds for a given component.
2. Student will be able to apply the knowledge of arc welding to join two metal pieces.
3. Student will be able to practice plain turning, facing, step turning, taper turning, and thread cutting operations on the lathe machine.
4. Student will be able to generate horizontal, vertical and angular surfaces on a given work piece using shaper.
5. Student will be able to generate spur gear on milling machine.
6. Student will be able to demonstrate Capstan and Turret lathe, cylindrical grinder and surface grinding machine.

LIST OF EXPERIMENTS

1. Use of basic tools and operations of the following trades

S.No.	Trade/Machine	No. of exercises
1	Moulding	3
2	Welding	3
3	Lathe Machine	3
4	Milling Machine	1
5	Shaping Machine	1

2. Moulding sand testing (Not for examination only for demonstration purpose)
3. Forging (Not for examination only for demonstration purpose)

Reference Books:

1. Elements of Workshop Technology Vol-1: Manufacturing Processes by S.K. Hajra Choudhury, A.K. HajraChoudhury, Nirjhar Roy, MPP, Pvt. Ltd.
2. Elements of Workshop Technology Vol-2: Machine Tools by S.K. Hajra Choudhury, A.K. HajraChoudhury, Nirjhar Roy, MPP, Pvt. Ltd.

STRENGTH OF MATERIALS LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int. Marks : 50
Ext. Marks : 50
Credits : 2

Course Outcomes:

1. Ability to identify different types of loads and measure them.
2. Ability to measure material properties of different materials using different methods.
3. Ability to measure bulking property and fineness of sand grains.

Course Objectives:

1. To understand the different types of loading and measure the loads.
2. To understand the material properties of different materials and the ways of finding them.
3. To understand the bulking property and fineness of sand grains and the methods of finding them.

LIST OF EXPERIMENTS

1. To study the stress strain characteristics (tension and compression) of metals by using UTM.
2. To study the stress strain characteristics of metals by using Hounsefield Tensometer.
3. Determination of compression strength of wood.
4. Determination of hardness using different hardness testing machines-Brinnels, Vickers and Rockwell's.
5. Impact test by using Izod and Charpy methods.
6. Deflection test on beams using UTM.
7. Tension shear test on M.S. Rods.
8. To find stiffness and modulus of rigidity by conducting compression tests on springs.
9. Torsion tests on circular shafts.
10. Bulking of sand.
11. Punch shear test, hardness test and compression test by using Hounsefield tensometer.
12. Sieve Analysis and determination of fineness number.

Reference Books:

1. Strength of Materials, by Timoshenko
2. Strength of Materials -By Jindal, Umesh Publications.
3. Strength of Materials by Andrew Pytel and Ferdinond L. Singer Longman.

Code: B17 ME 2209

INDUSTRY ORIENTED TECHNOLOGY LAB

Lab	: 2 Periods	Sessionals	: 50
Exam	: 3 Hrs.	Credits	: 1

CATIA

Course Objectives:

1. Increase ability to communicate with people
2. Learn to sketch and take field dimensions.
3. Learn to take data and transform it into graphic drawings.
4. Learn basic CATIA skills.
5. Learn basic engineering drawing formats
6. Prepare the student for future Engineering positions

Course Outcomes

Up on completion of the course the student shall be able to gain knowledge on:

1. CATIA screen and various Tool bars and menus and Explain about Dimensioning and Hatching
2. Draw the 2D – drawings like knuckle joint, screw jack, flange coupling, lathe tool post, eccentric etc.,
3. Explain about 3D solids and solids tool bar options and Drawing of 3D – components like bolt & nut, screw jack.
4. Rendering of 3D images.

LIST OF EXERCISES

1. Study the CATIA CAD screen, various toolbars and menus
2. Exercises on usage of Draw and modify tool bar.
3. Exercises on mirror, rotate, array and move commands
4. Exercises on dimension and hatching
5. Draw the 2D knuckle joint with full details & dimensioning
6. Draw the screw jack 2D drawing
7. Study the 3D solids (primitives) and solids tool bar options
8. Draw bolt and nut in 3D
9. Draw various parts of screw jack in assemble them as 3D component
10. Render the 3D images already generated and apply materials and light.

Reference Books:

1. **Lab Manual.**

ENGLISH PROFICIENCY-II
(Common to All Branches)

Lecture	: 1Period		
Tutorial	: 1 Periods	Int.Marks	: --
Exam	: ---	Credits	: --

AIM:

To equip the students with the components of the language required and help them gain adequate knowledge so as to become employable and competent.

Course Objectives:

Students will be able to

1. Enhance their Interpretative skills
2. Understand how to prepare a text.
3. Comprehend various types of writing discourses and respond.
4. Produce effectively different write-ups related to various business contexts.
5. Strengthen their emotional make-up.
6. Perceive various writing discourses.

Course Outcomes:

The students will

1. Develop the skills of taking and making notes
2. Interpret the pictures appropriately and effectively.
3. Read, comprehend and infer a given piece of writing effectively.
4. Learn and practice the skills of Research writing.
5. Communicate well through various forms of writing.
6. Be confident in giving presentations and dealing with people.

SYLLABUS

UNIT-1:SPEAKING

Analyzing proverbs

Enactment of One-act play

UNIT-2:READING

Reading Comprehension

Summarizing Newspaper Article

UNIT-3:WRITING

Note Taking & Note Making

Precis Writing

Essay Writing

Letter Writing

Picture Description

Literary Appreciation– Learning the Language of Literature

UNIT-4: VOCABULARY

Indian-origin English Words

Phrasal Verbs for Day-to-Day Communication

Commonly used Idiomatic Expressions

UNIT-5: PROJECT

Research Writing

Reference Books:

1. English for Colleges by Brendan J. Carroll, Macmillan Publications
2. Effective Technical Communication by M.AshrafRizwi. Tata Mcgraw Hill
3. Enriching Speaking and Writing Skills, Orient Blackswan Publishers
4. The Oxford Guide to Writing and Speaking by John Seely OUP
5. Six Weeks to Words of Power by Wilfred Funk. W.R.Goyal Publishers
6. English for Engineers and Scientists by Sangeeta Sharma and Binod Mishra, PHI Learning



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Affiliated to JNTUK, Kakinada), (Recognised by AICTE, New Delhi)

Accredited by NAAC with 'A' Grade

Recognised as Scientific and Industrial Research Organisation

CHINNA AMIRAM (P.O):: BHIMAVARAM :: W.G.Dt., A.P., INDIA :: PIN: 534 204

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

MECHANICAL ENGINEERING

I-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17 ME 3101	Operations Research	3	3	1	--	4	30	70	100
B17 ME 3102	IC Engines & Gas Turbines	3	3	1	--	4	30	70	100
B17 ME 3103	Kinematics of Machines	3	3	1	--	4	30	70	100
B17 ME 3104	Design of Machine Elements	3	3	1	--	4	30	70	100
B17 ME 3105	Fluid Machines & Systems	3	3	1	--	4	30	70	100
B17 ME 3106	Industrial Measurements & Metrology	3	3	1	--	4	30	70	100
B17 ME 3107	IC Engines & Kinematics of Machines Lab	2	--	--	3	3	50	50	100
B17 ME 3108	Metrology Lab	2	--	--	3	3	50	50	100
B17 BS 3101	Problem Solving & Linguistic Competence	1	--	3	--	3	30	70	100
B17 ME 3109	Modeling Lab	1	--	--	3	3	50	50	100
B17 BS 3105	IPR & Patents	--	--	2	--	2	--	---	--
Total		24	18	11	9	38	360	640	1000

OPERATIONS RESEARCH

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

This Course aims to:

1. Introduce students to use quantitative methods and techniques for effective Decisions–making; model formulation and applications that are used in solving business and industrial decision making problems.
2. Familiarize the students with the use of practice oriented mathematical applications for optimization functions in an organization.

Course Outcomes:

Upon completion of this course,

Students on an overall perspective will be able to:

1. Formulate a real time situation into a mathematical model.
2. Identify and develop operational research models from verbal description of real system.
3. Formulate simple reasoning, learning and optimization problems, in terms of the representations and methods presented.
4. Demonstrate the hand execution of basic reasoning and optimization algorithms on simple problems.
5. Formulate more complex, but still relatively simple problems, and apply implementations of selected algorithms to solve these problems.
6. Apply and analyze mathematical optimization functions to various applications.

Students will specifically able to:

1. Find the best use of an organization's resources.
2. Design an optimum distribution schedule of products from different sources to different destinations.
3. Allocate various resources to various activities on a one to one basis.
4. Assign a right job to a right person using job sequencing.
5. Design optimum schedules for projects.
6. Make right decisions and strategies in operations management using game theory and queuing theory.
7. Define optimum inventory policies suitable to a given situation.

SYLLABUS**UNIT-I**

Introduction to OR: Definition of OR, Characteristics and phases of OR, Scope of OR, OR Models, General Methods for Solving OR Models, Role of Computers in OR.

Linear Programming: Formulation, Graphical Solution, Simplex Method, Artificial Variable Technique – Big M method, Duality.

UNIT-II

Transportation Model: Mathematical Formulation, Tabular Representation, Balanced and unbalanced transportation Problems - Initial Solution by VAM and Optimality test by MODI Method, Degeneracy in TP.

Assignment Model: Mathematical Formulation, Hungarian Algorithm, Balanced and unbalanced Assignment Problems, Travelling Salesman Problem.

UNIT-III

Job Sequencing: Introduction, Assumptions, Johnson's Algorithm for Sequencing n - jobs through 2 machines, Problem with n- jobs and 3 machines, Problems with n - jobs and m - machines, Graphical Solution for 2 - Jobs and m - machines problem.

Inventory Models: Definition of inventory, costs associated with inventory problems, classification of inventory models, Deterministic inventory models - EOQ model without and with shortages, Production inventory model without and with shortages, Inventory models with price - breaks.

UNIT-IV

Games Theory: Introduction, Basic definitions, Two - Person zero - sum games - Minimax (maximin) criterion, saddle point, value of a game, Solution of games with Saddle point, Mixed strategy games - Arithmetic method for (2 x 2) games, Dominance principle to reduce size of game, solution of (2 x n) and (m x 2) games, Algebraic Solution to rectangular games.

Queuing Models: Structure of queuing models, characteristics of Queuing process, Kendall's notation, Single channel systems - (M/M/1:∞/FIFO) model and (M/M/1:N/FIFO) model.

UNIT-V

Network Analysis: Introduction, Project Scheduling by CPM and PERT, Network diagram representation, rules for drawing network diagram, Labeling by Fulkerson's rule, Network calculations - EST, EFT, LST, LFT, Float/Slack and critical path, PERT calculations.

Text Books:

1. Operations Research by S.D. Sharma (Kedarnadh & Ramnadh & Co., Meerut).
2. Operations Research by V.K.Kapoor (Sultan Chand & Sons, New Delhi).

Reference Books:

1. Operations Research by H.A. Taha (PHI, New Delhi).
2. Operations Research: Methods and Problems by M. Sasiene, A. Yespal and L. Friedman (John Wiley & Sons, New York).
3. Operations Research by Kanti Swaroop, Gupta, Manmohan (Sultan Chand & Sons, New Delhi)

IC ENGINES & GAS TURBINES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To impart the knowledge of basic gas power cycles and combustion phenomena, various performance factors and design aspects of SI and CI Engines.
2. To impart the knowledge of combustion phenomenon taking place in SI engines
3. To impart the knowledge of combustion phenomenon taking place in CI engines
4. To impart the knowledge of Reciprocating compressors and Rotary compressors methods to improve their efficiency.
5. To impart the knowledge of gas turbine cycles, and their components and efficiency improving techniques, and jet propulsion.

Course Outcomes:

1. Apply the knowledge of gas power cycles adequately and can calculate their efficiencies.
2. Explain the processes involved in combustion in S.I Engines.
3. Explain the processes involved in combustion in C.I Engines.
4. Apply the knowledge of reciprocating and rotary compressors in engineering applications.
5. Compute and develop various methods to improve the efficiency of gas turbine power plants, and can explain jet propulsions.

SYLLABUS

UNIT-I

I.C. engines: classification-comparison of two stroke and four stroke engines- comparison of S.I. and C.I. engines-Valve timing and port timing diagrams- Efficiencies- air standard efficiency, indicated thermal efficiency, brake thermal efficiency, mechanical efficiency, volumetric efficiency and relative efficiency-Testing and performances of I.C. engines -Basic principles of carburetion and fuel injection.

UNIT-II

Combustion in S.I. Engines- Normal combustion and abnormal combustion Importance of flame speed and effect of engine variables-types of abnormal combustion pre-ignition and knock, Fuel requirements and fuel rating, anti-knock additions- Combustion chamber requirements and Types of combustion chamber- Design principles of combustion chambers.

UNIT-III

Combustion in C.I. Engines- Stages of combustion- Delay period and its importance- effect of engine variables, diesel knock, suction compression and combustion induced turbulence, open and divided combustion chambers.

UNIT-IV

Reciprocating and Rotary Compressors: Reciprocating compressors-effect of clearance in compressors, volumetric efficiency-single stage and multi stage compressors-effect of inter cooling in multi stage compressors-Vane type blower-centrifugal compressor- Adiabatic efficiency- Diffuser- Axial flow compressors- Velocity diagrams, degree of reaction, performance characteristics.

UNIT-V

Gas Turbines: Simple gas turbine plant- Ideal cycle, closed cycle and open cycle for gas turbines Efficiency, work ratio and optimum pressure ratio for simple gas turbine cycle- Parameters of performance- Actual cycle, regeneration, Inter-cooling and reheating, closed and semi-closed cycle Jet propulsion and Rockets.

Text Books:

1. A Treatise on Heat Engineering by Vasandhani and Kumar.
2. Applied Thermodynamics-II by R. Yada

Reference Books:

1. I.C. Engines by V. Ganesan.
2. Thermal Engineering, by R.K.Rajput.
3. I.C. Engines, by Mathur and Nehata.
4. Gas Turbines, by Cohen and Rogers.

KINEMATICS OF MACHINES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. The objective is to study the relative motion, velocity, and accelerations of the various elements in a mechanism.
2. In mechanical Engineering we come across number of mechanisms such as four bar/slider crank/double slider crank/straight line motion mechanism etc. Mechanism deals with only relative motions. Once we make a study considering for us also there it is called kinetics.
3. The first course deals with mechanisms, their inversions straight line motion mechanisms steering mechanisms etc.
4. The study of cams/gears & gear trains.

Course Outcomes:

1. Understand the basic principles of mechanisms in mechanical engineering applications.
2. Understand the mechanisms, their inversions straight line motion mechanisms steering mechanisms etc.
3. Understand the importance of toothed gears, gear trains.
4. Understand the cam their practice application.
5. Understand the importance of relative motion, velocity, and accelerations of the various elements in a mechanism

SYLLABUS**UNIT-I**

Mechanisms: Elements or Links – Classification – Rigid Link, flexible and fluid link – Types of kinematics pairs – sliding, turning, rolling, screw and spherical pairs – lower and higher pairs – closed and open pairs – constrained motion – completely, partially or successfully and incompletely constrained .

Mechanism and Machines – Mobility of Mechanisms : Grubler's criterion, classification of machines – kinematics chain – inversions of mechanism – inversions of four bar chain, single and double slider crank chains, Mechanical Advantage

UNIT-II

Kinematics: Velocity and acceleration – Motion of link in machine – Determination of Velocity and acceleration – Graphical method – Application of relative velocity method.

Plane motion of body: Instantaneous center of rotation– Three centers in line theorem – Graphical determination of instantaneous center, determination of angular velocity of links by instantaneous center method.

Analysis of Mechanisms: Analysis of slider crank chain for displacement- velocity and acceleration of slider – Acceleration diagram for a given mechanism.

Kliens construction - Coriolis acceleration - determination of Coriolis component of acceleration

UNIT-III

Straight-line motion mechanisms: Exact and approximate copied and generated types – Peaucellier - Hart - Scott Russel- Modified Scott Russel – Grasshopper – Watt -Tchebicheff's - Pantographs

Steering gears: Conditions for correct steering – Davis Steering gear, Ackerman's steering gear.

Hooke's Joint: Single and double Hooke's joint –velocity ratio – application – problems

UNIT-IV

Cams: Definitions of cam and followers – their uses – Types of followers and cams – Terminology – Types of follower motion - Uniform velocity, Simple harmonic motion and uniform acceleration and retardation. Construction of cam profiles- Cam with knife edged follower and roller follower Maximum velocity and maximum acceleration during outward and return strokes.

UNIT-V

Higher pair: Friction wheels and toothed gears – types – law of gearing, condition for constant velocity ratio for transmission of motion – involutes profiles – phenomena of interferences – Methods of interference. Condition for minimum number of teeth to avoid interference – expressions for arc of contact and path of contact of Pinion & Gear

Gear Trains: Introduction – Types – Simple – compound and reverted gear trains – Epicyclic gear train. Methods of finding train value or velocity ratio of Epicyclic gear trains. Torques in epicyclic gear trains

Text Books:

1. Theory of Machines by R.S.Khurmi & J.K.Gupta
2. Theory of Machines / S.S.Rattan / Mc Graw Hill Publishers

Reference Books:

1. Theory of Machines by Thomas Bevan
2. Theory of Machines and Mechanisms/JOSEPH E. SHIGLEY/ Oxford

DESIGN OF MACHINE ELEMENTS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Students will be acquainted with standards like ASTM, ASME etc., safety, reliability, importance of dimensional parameters, manufacturing aspects in mechanical design.
2. Students will understand to formulate and analyze stresses and strains in machine elements like shafts, springs etc and structures under static and/or dynamic load conditions.

Course Outcomes:

On the successful completion of the course, students will be able to:

1. Explain the design concepts of static strength of mechanical components
2. Explain the design concepts of fatigue strength of mechanical components
3. Determine the strength of the threaded and welded joints
4. Design the shafts, rigid and flexible couplings parametrically for different loading conditions.
5. Design the energy absorbing mechanical components such as springs for the specified loading conditions.

SYLLABUS**UNIT-I**

Introduction: Machine Design, basic procedure of machine design, traditional design methods, general design models, BIS system of designation of steels, manufacturing considerations in design.

Design against static loads: Modes of failure, Factor of safety, stress strain relationship, shear stress and shear strain, stresses due to bending moment, stresses due to torsional moment, eccentric axial loading, Design of cotter and knuckle joints, Static failure theories.

UNIT-II

Design Against Fluctuating Load: stress concentration, stress concentration factors, reduction of stress concentration, fluctuating stresses, fatigue failure, endurance limit, notch sensitivity, Soderberg, Goodman and modified Goodman diagrams, Gerber equation, fatigue design under combined stresses.

UNIT-III

Threaded joints: forms of threads, basic types of screw fastenings, ISO metric screw threads, eccentrically loaded bolted joints.

Welded joints: different types welded joints and their design aspects, Types and strength of weld joints subjected to bending loads and eccentric loads.

UNIT-IV

Shafts: Shafts design on strength basis, torsional rigidity basis, Design of hollow shafts, flexible shafts, ASME codes for shafts,

Keys: keys design, Flat & square keys; **Couplings:** Rigid and flange couplings, Flexible couplings.

UNIT-V

Spring Design: classification and spring materials, Design of helical compression springs and helical springs subjected to fatigue loading, Design of laminated springs, Equalized stress in spring leaves, Surge in springs, Nipping and shot peening.

Text Books:

1. Design of Machine Elements by V.B.Bhandari, TMH Publishing Co. Ltd., New Delhi.
2. Machine Design by Jain, Khanna Publications.

Reference Books:

1. Machine Design by Pandya and Shaw, Charotar publications.
2. Machine design, an integrated approach by R.L.Norton, 2nd edition, Pearson Education

FLUID MACHINES & SYSTEMS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To Study about Impact of jets and basics of jet propulsion.
2. Understand the concepts of turbines, types, draft tube, specific speed and unit quantities.
3. To study about Centrifugal pumps, Multi stage pumps, Priming and Cavitation.
4. To study about Reciprocating pumps, Separation and Air vessels.
5. To study and understand various types of hydraulic devices.

Course Outcomes:

After the completion of the course, students are able to

1. Understand the concepts of jets and jet propulsion and its applications in fluid machinery.
2. Gain the knowledge such as workdone, specific speed, performance curves and governing of impulse and reaction turbines.
3. Understand the centrifugal pumps - Multi stage pumps, Minimum speed required to start the pump, Performance curves.
4. Understand the various aspects of Reciprocating pumps such as working, indicator diagram, airvessels.
5. Understand description and working of various types of hydraulic devices.

SYLLABUS**UNIT-I**

Impact of jet and jet propulsion: Impact of jet on stationary surfaces- Impact of jet on hinged surfaces- moving curved vane with tangential entry of water- Radial flow over the vanes- Jet propulsion of tank and ships

UNIT-II

Hydraulic Turbines: Classification- Pelton wheel- Reaction turbines- Inward and outward radial flow reaction turbines- Francis turbine- Axial flow reaction turbine- Kaplan turbine- Draft tube- Types- Theory- and efficiency of draft tube.

Specific Speed: Determination- Significance- Unit quantities- Unit speed- Unit discharge and unit power- Characteristic curves of hydraulic turbines- Constant head curves- Constant speed curves and Iso-efficiency curves- Governing of turbines.

UNIT-III

Centrifugal Pumps: Main parts- Efficiency- Minimum speed for starting- Multi-stage centrifugal pumps- Specific speed of a centrifugal pump- Priming of a centrifugal pump- Characteristic curves- Main, Operational and constant efficiency curves- Cavitation- Effects- Cavitation in Hydraulic machines.

UNIT-IV

Reciprocating Pumps: Main parts- Classification- Velocity and acceleration variation in suction and delivery pipes due to piston acceleration- Effect of variation of velocity on friction in suction and delivery pipes- Effect of acceleration in suction and delivery pipes on indicator diagram- Effect of friction- Maximum speed of reciprocating pump- Airvessels, Simple problems on air vessels.

UNIT-V

Hydraulic Press- Hydraulic accumulator- Differential hydraulic accumulator- Hydraulic intensifier- Hydraulic ram- Hydraulic lift- Hydraulic crane- Fluid coupling- Hydraulic torque converter, Air lift pump.

Text Books:

1. Fluid Mechanics and Hydraulic Machinery, by R.K.Bansal, Laxmi Publications.
2. Fluid Mechanics and Hydraulic Machinery, by R.K.RAJPUT.

Reference Books:

1. Fluid Flow Machines, by N.S.Govinda Rao, Tata McGraw Hill publishing company Ltd.
2. Hydraulics & Fluid Mechanics, P.N. Modi and S.M. Seth, TEXTBOOKS House, Delhi.

INDUSTRIAL MEASUREMENTS & METROLOGY

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The main objectives of this module are:

1. To provide to the students an understanding and appreciation of the science of Measurement.
2. To expose the students to various mechanical and electrical engineering measuring devices, and understand the different degree of accuracy obtained from different types of instruments.

Course Outcomes:

Students who successfully complete this course will be able to:

1. identify the uncertainties in dimensional metrology and the define the measurement standards;
2. describe the fundamentals of dimensional and geometrical tolerances;
3. measure length and angles using line-graduated instruments, i.e. Vernier calipers, micrometers, bevel protractor, sine bar and surface plates;
4. use comparative length-measuring instruments, i.e. dial indicator, to measure variations in the distance between two or more surfaces;
5. use effective methods of measuring straightness, flatness, roundness, profile, screw threads and gear teeth;
6. measure dimensions of shafts, bearings and linear surfaces in metric and imperial units using calibers, micrometers, and scales;
7. use contour projector and coordinate measuring machine to record measurements of complex profiles with high sensitivity;
8. Use gage blocks, fixed gages, pneumatic gages gage blocks to measure various work pieces.
9. explain the effect of environmental conditions on the accuracy of measurements;
10. demonstrate the correct methods for adjustment and calibration of various measuring instruments;
11. Use appropriate method for determination of accuracy based on product function and manufacturing capability.

SYLLABUS**UNIT-I**

Instrumentation: Concepts of measurements, static performance characteristics, accuracy of measurement and its analysis. Instrumentation for measurement: Force(analytical balance, proving rings), torque(dynamometers), strain(opto-mechanical method),Pressure(Mc-leodguage), flow(venturimeter, orifice meter, nozzel), Temperature (bimetallic thermostat, liquid- in-glass, optical)and vibration(optical).

UNIT-II

Optical Methods of Measurement: Introduction, Laserbeam as a light pointer, length/displacement measurement, temperature sensors, seismographic measurement. Introduction to fiber optics, fiber types, properties of optical fibres and a fibre optic sensor configuration.

UNIT-III

limits, Fits and gauges: limits, fits and Interchangeability, Plain limit gauges, Measurement of screw threads:major diameters, Minor diameters and effective diameter, Pitch. Limit gauges for internal and external threads. Measurement of spur gears: pitch, profile, lead, backlash, tooth thickness.

UNIT-IV

Measuring devices:Tool maker's microscope, Slip gauges, comparator: Twisted strip type, Optical lever, Electric, Pneumatic,Optical projector. Optical bevel protractor, Sine bar, Angle gauges, Precision level, Autocollimator, Angle dekkor, Optical dividing heads and rotary tables, Straightness measurement,Squareness testing,Flatness measurement, Roundness measurement.

UNIT-V

Surface texture and Acceptance tests:Surface textureParameters, sampling length, Specification, Stylus instruments for surface roughness measurement. Acceptance tests on machine tools: Lathe, Milling machine and Radial drilling machine.

Text Books:

1. Process & Materials of Manufacture, R.A.Lindberg, 4th edition, Prentice-Hall of India, New Delhi.
2. A Text Book of Engineering Metrology, I.C.Gupta, DhanpatRai& Sons, Delhi.
3. Mechanical Measurements, by R.S.Sirohi, H.G. Radha Krishna, Wiley Eastern, New Delhi.

Reference Books:

1. A.S.T.M.E., Hand book of Industrial Metrology, Prentice-Hall of India, New Delhi.
2. A.S.T.M.E., Hand book of Manufacturing Engineering.
3. Manufacturing Processes & Materials for Engineers, L.E.Doyle& others, Prentice-Hall of India, New Delhi.
4. Manufacturing Technology by Adithan, New age international.
5. Instrumentation for Engineering Measurements, by R.H. Cerni and L.E.Foster, J.Wiley& Sons, New York.

IC ENGINES & KINEMATICS OF MACHINES LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int. Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. Students would appreciate the fundamentals of thermodynamics being extended to real time applications
2. Students might come out with innovative ideas which may be extended in the form of projects
3. Students by themselves should be able to trouble shoot basic essential issues of cars

Course Outcomes:

1. A This course comprehensively deals with practical approach to I.C Engines and four bar chain mechanisms
2. To expose students to different methods of finding friction power in single and multi cylinder engines
3. Understand the working of Gyroscopes
4. Deal with basic trouble shooting aspects and specifications of Car model: Maruthi ZEN

LIST OF EXPERIMENTS

(a) IC Engines Lab

1. Load test on single cylinder diesel Engine.
2. Morse test on multi-cylinder petrol engine.
3. Heat balance sheet on I.C. Engines.
4. Study of multi-cylinder engines and determination of its firing order
5. Performance test on multi cylinder diesel engine
6. To rectify basic essential issues of modern cars such as Jump starting, Tyre replacement & repair, dismantling some components to understand their mechanisms etc.

(b) Kinematics of Machines Lab

1. Study of automobile mechanisms
2. Verification of laws of balancing.
3. Determination of ratios of angular speeds of shafts connected by Hooke's joint.
4. Determination of the ratio of times and ram velocities of Whitworth quick return motion mechanism.
5. To draw curves of slider displacement and crank angle and linear velocities w.r.t. time for a slider crank mechanism and compare with theoretical values.
6. To determine the relation of gyroscopic couple and compare with the theoretical values.

Reference Books:

1. Thermal Engineering, by R.K.Rajput.
2. I.C. Engines, by Mathur and Nehata.

METROLOGY LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int. Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To demonstrate the fundamentals of Metrology and Measurement Engineering concepts, different tools used for minute measurements, and calibrate various kinds of measuring instruments.
2. To demonstrate the techniques and processes followed for calibrating measuring instruments like Micrometer, Mechanical Comparator, and Vernier Caliper etc. and educate students about the metrology equipment.

Course Outcomes:

1. Students will understand construction and working of various measuring instruments and its calibration.
2. The student will be able to operate measurement instruments on their own and test different components for their dimensional accuracy.
3. Students will be able to understand application of gauges.

LIST OF EXPERIMENTS

1. Calibration of vernier calipers.
2. Calibration of outside micrometer.
3. Calibration of tool room microscope.
4. Calibration of mechanical comparator.
5. Measurement of taper using dial gauge and sine bar.
6. Study and use of bevel protractor.
7. Measurement of the height of circular spigot.
8. Measurement of angle of v-groove.
9. Measurement of distance between two holes of a template using vernier height gauge.
10. Measurement of gear parameters.
11. Establishing thread details on profile projector.
12. Measurement of tool angles on profile projector.

Reference Books:

1. A Text Book of Engineering Metrology, I.C.Gupta, Dhanpat Rai & Sons, Delhi.
2. Mechanical Measurements, by R.S.Sirohi, H.G. Radha Krishna, Wiley Eastern, New Delhi.
3. A.S.T.M.E., Hand book of Industrial Metrology, Prentice-Hall of India, New Delhi.
4. A.S.T.M.E., Hand book of Manufacturing Engineering.
5. Manufacturing Processes & Materials for Engineers, L.E.Doyle & others, Prentice-Hall of India, New Delhi.
6. Manufacturing Technology by Adithan, New age international.
7. Instrumentation for Engineering Measurements, by R.H. Cerni and L.E.Foster, J.Wiley & Sons, New York.

PROBLEM SOLVING & LINGUISTIC COMPETENCE**(Common to all Branches)**

Tutorial : 3 Periods (VA-2+QA-1)
Exam : 3 Hrs.

Int.Marks : 30
Ext.Marks : 70
Credits : 1

Part-A: Verbal and Soft Skills-I**Course Objectives:**

1. To introduce concepts required in framing grammatically correct sentences and identifying errors while using Standard English.
2. To familiarize the learner with high frequency words as they would be used in their professional career.
3. To inculcate logical thinking in order to frame and use data as per the requirement.
4. To acquaint the learner of making a coherent and cohesive sentences and paragraphs for composing a written discourse.
5. To familiarize students with soft skills and how it influences their professional growth.

Course Outcomes:

The student will be able to

1. Detect grammatical errors in the text/sentences and rectify them while answering their competitive/ company specific tests and frame grammatically correct sentences while writing.
2. Answer questions on synonyms, antonyms and other vocabulary based exercises while attempting CAT, GRE, GATE and other related tests.
3. Use their logical thinking ability and solve questions related to analogy, syllogisms and other reasoning based exercises.
4. Choose the appropriate word/s/phrases suitable to the given context in order to make the sentence/paragraph coherent.
5. Apply soft skills in the work place and build better personal and professional relationships making informed decisions.

SYLLABUS**Grammar: (VA)**

Parts of speech(with emphasis on appropriate prepositions, co-relative conjunctions, pronouns-number and person, relative pronouns), articles(nuances while using definite and indefinite articles), tenses(with emphasis on appropriate usage according to the situation), subject – verb agreement (to differentiate between number and person) , clauses(use of the appropriate clause , conditional and relative clauses), phrases(use of the phrases, phrasal verbs) to-infinitives, gerunds, question tags, voice, direct & indirect speech, degrees of comparison, modifiers, determiners, identifying errors in a given sentence, correcting errors in sentences.

Vocabulary: (VA)

Synonyms and synonym variants(with emphasis on high frequency words), antonyms and antonym variants(with emphasis on high frequency words), contextual meanings with regard to inflections of a word, frequently confused words, words often mis-used, multiple meanings of the same word (differentiating between meanings with the help of the given context), foreign phrases, homonyms, idioms, pictorial representation of words, word roots, collocations.

Reasoning: (VA)

Critical reasoning (understanding the terminology used in CR- premise, assumption, inference, conclusion), Analogies (building relationships between a pair of words and then identifying similar relationships), Sequencing of sentences (to form a coherent paragraph, to construct a meaningful and grammatically correct sentence using the jumbled text), odd man (to use logical reasoning and eliminate the unrelated word from a group), YES-NO statements (sticking to a particular line of reasoning Syllogisms).

Usage: (VA)

Sentence completion (with emphasis on signpost words and structure of a sentence), supplying a suitable beginning/ending/middle sentence to make the paragraph coherent, idiomatic language (with emphasis on business communication), punctuation depending on the meaning of the sentence.

Soft Skills:

Introduction to Soft Skills – Significance of Inter & Intra-Personal Communication – SWOT Analysis –Creativity & Problem Solving – Leadership & Team Work - Presentation Skills Attitude – Significance – Building a positive attitude – Goal Setting – Guidelines for Goal Setting – Social Consciousness and Social Entrepreneurship – Emotional Intelligence - Stress Management, CV Making and CV Review.

Text Books:

1. Oxford Learners's Grammar – Finder by John Eastwood, Oxford Publication.
2. R S Agarwal's books on objective English and verbal reasoning
3. English Vocabulary in Use- Advanced , Cambridge University Press.
4. Collocations In Use, Cambridge University Press.
5. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
6. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. English Grammar in Use by Raymond Murphy, CUP
2. Websites: Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. The Art of Public Speaking by Dale Carnegie
5. The Leader in You by Dale Carnegie
6. Emotional Intelligence by Daniel Golman
7. Stay Hungry Stay Foolish by Rashmi Bansal
8. I have a Dream by Rashmi Bansal

Part-B: Quantitative Aptitude -I

Course objectives:

The objective of introducing quantitative aptitude-1 is:

1. To familiarize students with basic problems on numbers and ratio's problems.
2. To enrich the skills of solving problems on time, work, speed, distance and also measurement of units.
3. To enable the students to work efficiently on percentage values related to shares, profit and loss problems.
4. To inculcate logical thinking by exposing the students to reasoning related questions.
5. To expose them to the practice of syllogisms and help them make right conclusions.

Course Outcomes:

1. The students will be able to perform well in calculating on number problems and various units of ratio concepts.
2. Accurate solving problems on time and distance and units related solutions.
3. The students will become adept in solving problems related to profit and loss, in specific, quantitative ability.
4. The students will present themselves well in the recruitment process using analytical and logical skills which he or she developed during the course as they are very important for any person to be placed in the industry.
5. The students will learn to apply Logical thinking to the problems of syllogisms and be able to effectively attempt competitive examinations like CAT, GRE, GATE for further studies.

SYLLABUS

Numbers, LCM and HCF, Chain Rule, Ratio and Proportion Importance of different types of numbers and uses of them: Divisibility tests, Finding remainders in various cases, Problems related to numbers, Methods to find LCM, Methods to find HCF, applications of LCM, HCF. Importance of chain rule, Problems on chain rule, Introducing the concept of ratio in three different methods, Problems related to Ratio and Proportion.

Time and work, Time and Distance Problems on man power and time related to work, Problems on alternate days, Problems on hours of working related to clock, Problems on pipes and cistern, Problems on combination of the some or all the above, Introduction of time and distance, Problems on average speed, Problems on Relative speed, Problems on trains, Problems on boats and streams, Problems on circular tracks, Problems on polygonal tracks, Problems on races.

Percentages, Profit Loss and Discount, Simple interest, Compound Interest, Partnerships, shares and dividends

Problems on percentages-Understanding of cost price, selling price, marked price, discount, percentage of profit, percentage of loss, percentage of discount, Problems on cost price, selling price, marked price, discount. Introduction of simple interest, Introduction of compound interest, Relation between simple interest and compound interest, Introduction of partnership, Sleeping partner concept and problems, Problems on shares and dividends, and stocks.

Introduction, number series, number analogy, classification, Letter series, ranking, directions Problems of how to find the next number in the series, Finding the missing number and related sums, Analogy, Sums related to number analogy, Ranking of alphabet, Sums related to Classification, Sums related to letter series, Relation between number series and letter series, Usage of directions north, south, east, west, Problems related to directions north, south, east, west.

Data sufficiency, Syllogisms Easy sums to understand data sufficiency, Frequent mistakes while doing data sufficiency, Syllogisms Problems.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi

References:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

Websites:

1. www.m4maths.com
2. www.Indiabix.com
3. www.800score.com
4. Official GRE site
5. Official GMAT site

MODELLING LAB

Lab : 3 Periods
Exam : 3 Hrs.

Int.Marks : 50
Ext. Marks : 50
Credits : 1

Course Objectives:

1. Increase ability to communicate with people
2. Learn to sketch and take field dimensions.
3. Learn to take data and transform it into graphic drawings.
4. Learn basic modelling skills.
5. Learn basic engineering drawing formats
6. Prepare the student for future Engineering positions

Course Outcomes:

Up on completion of the course the student shall be able to gain knowledge on:

1. Modelling screen and various Tool bars and menus and Explain about Dimensioning and Hatching
2. Draw the 2D – drawings like knuckle joint, screw jack, flange coupling, lathe tool post, eccentric etc.
3. Explain about 3D solids and solids tool bar options and Drawing of 3D – components like bolt & nut, screw jack.
4. Rendering of 3D images.

LIST OF EXERCISES

1. Study the Modelling screen, various toolbars and menus
2. Exercises on usage of Draw and modify tool bar.
3. Exercises on mirror, rotate, array and move commands
4. Exercises on dimension and hatching
5. Draw the 2D knuckle joint with full details & dimensioning
6. Draw the screw jack 2D drawing
7. Study the 3D solids (primitives) and solids tool bar options
8. Draw bolt and nut in 3D
9. Draw various parts of screw jack in assemble them as 3D component
10. Render the 3D images already generated and apply materials and light.

Reference Books:

1. Lab Manual.

IPR & PATENTS
(Common to CE, EEE & ME)

Tutorial : 2 Periods

Credits : 0

Course Objectives:

1. To introduce the idea of tangible and intangible property and its protection.
2. To familiarize with the frameworks for protection of intellectual property.
3. To layout the procedures to claim rights over intellectual property.

Course Outcomes:

After successful completion of the course, the student shall be able to

1. Identify various types of intangible property that an engineering professional could generate in the course of his career.
2. Distinguish between various types of protection granted to Intellectual Property such as Patents, Copy Rights, Trademarks etc.,
3. List the steps involved in getting protection over various types of intellectual property and maintaining them.
4. Take precautions in writing scientific and technical reports without plagiarism.
5. Help micro, small and medium entrepreneurs in protecting their IP and respecting others IP as part of their business processes.

SYLLABUS

UNIT I

Intellectual Property Law: Basics - Types of Intellectual Property - Innovations and Inventions - Trade related Intellectual Property Rights – Agencies Responsible for Intellectual Property Registration – Infringement - Compliance and Liability Issues

UNIT II

Principles of Copyright – Subject Matters of Copyright – Rights Afforded by Copyright Law – Copyright Ownership– Copyright Formalities and Registration – Limitations – Infringement of Copyright - Plagiarism and difference between Copyright infringement and Plagiarism

UNIT III

Introduction to Trade Mark – Trade Mark Registration Process – Post registration procedures – Trade Mark maintenance – Infringement – Dilution of Ownership of Trade Mark – Likelihood of confusion – Trade Mark claims – Trade Marks Litigation – International Trade Mark Law

UNIT IV

Introduction to Patent Law – Rights and Limitations – Rights under Patent Law – Patent Requirements – Ownership and Transfer – Patent Application Process and Granting of Patent – Patent Infringement and Litigation – International Patent Law – Double Patenting

UNIT V

Introduction to Trade Secrets – Maintaining Trade Secret – Physical Security – Employee Access Limitation – Employee Confidentiality Agreement – Trade Secret Law – Unfair Competition – Trade Secret Litigation – Breach of Contract – Applying State Law.

Text Books:

1. Kompal Bansal & Parikshit Bansal "Fundamentals of Intellectual Property for Engineers", BS Publications
2. Prabhuddha Ganguli: "Intellectual Property Rights" Tata McGraw – Hill, New Delhi
3. R. Radha Krishnan, S. Balasubramanian: "Intellectual Property Rights: Text and Cases", Excel Books, New Delhi.

Reference Books:

1. Deborah E. Bouchoux: "Intellectual Property". Cengage learning, New Delhi
2. Richard Stim: "Intellectual Property", Cengage Learning, New Delhi.

SCHEME OF INSTRUCTION & EXAMINATION

(Regulation R17)

III/IV B.TECH

(With effect from 2017-2018 Admitted Batch onwards)

MECHANICAL ENGINEERING

II-SEMESTER

Code No.	Name of the Subject	Credits	Lecture Hrs	Tutorial Hrs	Lab Hrs	Contact Hrs/Week	Internal Marks	External Marks	Total Marks
B17 ME 3201	Industrial Engineering & Management	3	3	1	--	4	30	70	100
B17 ME 3202	Control Systems	3	3	1	--	4	30	70	100
B17 ME 3203	Dynamics of Machines	3	3	1	--	4	30	70	100
B17 ME 3204	Machine Design	3	3	1	--	4	30	70	100
# OE	Open Elective	3	3	1	--	4	30	70	100
B17 ME 3207	Computer Aided Design	3	3	1	--	4	30	70	100
B17 ME 3208	Industrial Engineering Lab	2	--	--	3	3	50	50	100
B17 ME 3209	Fluid Mechanics and Machinery Lab	2	--	--	3	3	50	50	100
B17 BS 3201	Employability Skills	1	--	3	--	3	30	70	100
B17 BS 3202	Basic Coding	1	--	--	3	3	50	50	100
Total		24	18	9	9	36	360	640	1000

# OE (Open Elective)	B17 BS 3207	Entrepreneurship
	B17 CS 3213	Database Management System
	B17 CE 3207	Waste Water Management
	B17 CS 3210	Computer Graphics
	B17 ME 3205	Industrial Robotics
	B17 ME 3206	Green Engineering Systems

INDUSTRIAL ENGINEERING & MANAGEMENT

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Students learn

1. To know management practice in industry
2. To acquire capacity to handle industrial disputes
3. To know much about production activities and to improve productivity
4. To learn the work study procedures and quality concepts to get more productivity
5. To have exposure on some maintenance practices in industry

Course Outcomes:

Students will be able to

1. Apply management theories in industry
2. Know personnel management techniques to motivate the workers
3. Settle the industrial disputes in the organization
4. Acquire full knowledge on production planning and control procedures
5. Understand the economics of plant layout
6. Aware of material handling principles and equipment
7. Apply maintenance practices
8. Have knowledge on materials management
9. Improve the productivity by applying work study procedures and quality concepts

SYLLABUS**UNIT-I**

Concepts of Industrial Management: Principles of Management, Growth of management thought, Functions of management, Principles of organization, Types of organization and committees.

Personnel management and Industrial relations: Functions of personnel management, Theories of motivation, Discipline in industry, Promotion, Transfer, Lay off and Discharge, Labor turnover, Trade unions, Industrial disputes, Strikes, Lock-out, Picketing, Gherao, Settlement of industrial disputes, Collective bargaining, Industrial dispute act 1947 and Factories act 1948.

UNIT-II

Production planning and control: Types of productions, Production cycle, Product design and development, Process planning, Forecasting, Loading, Scheduling, Dispatching, Routing, Progress control, Line of Balance. Simple problems.

UNIT-III

Plant location & Plant layout: Economics of plant location, Rural Vs Suburban sites, Types of layouts, Types of buildings, Methods of plant layouts (Travel Chart Technique), the concept of Assembly line balancing. Simple problems.

Materials handling: Principles, Concept of unit load, Palletization and Containerization, Selection of material handling equipment, Types and applications of material handling equipment.

UNIT-IV

Work study: Concept of productivity, Method study – Basic steps in method study, Process charts, Diagrams, Models and Templates, Principles of motion economy, Micro motion study, Therbligs, SIMO chart, Work measurement – Stop watch procedure of time study, Performance rating, Allowances, Work sampling, Simple problems.

UNIT-V

Materials management: Introduction, Purchasing, Objectives of purchasing department, Buying techniques, Purchase procedure, Stores and material control, receipt and issue of materials, Store records, Inventory control, EOQ model (simple problems).

Quality control: Control charts of variables and attributes. Single and double sampling plans.

Text Book:

1. Industrial Engineering and Management by Dr. O P Khanna

Reference Books:

1. Principles of Management by Koontz & Donnel.
2. Production and Operations Management by Everette Adam & Ronald Ebert.
3. Operations Management by John McClain & Joseph Thames.
4. Industrial Engineering and Production Management by Martand Telsang

CONTROL SYSTEMS

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To introduce the basic principles of control systems to develop mathematical models for physical systems.
2. To familiarize the students on the basic concepts of feedback characteristics of control systems for standard test signals.
3. To familiarize the students on finding stability of control systems using time and frequency domain techniques.

Course Outcomes:

Students will be able to

1. Classify control systems and explain the needs and effects of feedback in a control system and Compute transfer function of multiple subsystems modeled as block diagram/ signal flow graph.
2. Develop mathematical models for physical systems using the knowledge of fundamental principles of mathematics and control systems.
3. Compute transfer function of multiple subsystems modeled as state space representation.
4. Compute and describe the output response and steady state error of first, second and higher order control systems for standard input signals
5. Determine the stability of a system using Routh Hurwitz and Nyquist criterion.

SYLLABUS**UNIT-I**

Introduction: Control systems, Classification of Control systems, Feedback and its effects. Transfer Function, Block Diagram and Signal Flow Graphs

UNIT-II

Mathematical Modelling of Physical Systems: Modelling of mechanical and electrical system elements, Equations of mechanical and electrical systems, Electrical analogous of mechanical systems.

UNIT-III

State-variable analysis: State variables, Matrix representation of state equations, State-Transition Matrix, State-Transition Equation, Relationship between state equations and high order differential equations, Relationship between state equations and transfer functions, Characteristic equation, eigen values and eigen vectors.

UNIT-IV

Time Response Analysis: Time response, Typical test signals for the time response of control systems, Order of a system, response of first and second order systems for various inputs, Time domain specifications, Type number of control systems, Steady state error, Static error constants.

UNIT-V

Frequency Response Analysis: Frequency response, Frequency-domain Analysis of Control Systems: Gain margin, Phase margin.

Stability of control systems: Stability, Characteristic equation, Methods of determining stability of linear control systems, Routh- Hurwitz criterion, Nyquist stability criterion, Application of the Nyquist criterion.

Text Books:

1. Control Systems by A. Nagoor Kani, RBA Publications.
2. Automatic Control Systems by Benjamin C Kuo
3. Advanced Control Theory by A. Nagoor Kani, RBA Publications.

Reference Books:

1. Control Systems Engineering by Nagrath/Gopal , New Age International.
2. Control systems principles and design by M Gopal, Tata Mcgraw-Hill.
3. Control systems A K Jairath, CBS problems and solutions series.

DYNAMICS OF MACHINES

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the concept of turning moment diagrams, flywheel and speed control systems such as governors.
2. To understand the concept of bearings, clutches and effect of friction in motion transmission and in machine components.
3. Develop knowledge of analytical and graphical methods for calculating balancing of rotary and reciprocating masses.
4. Develop understanding of vibrations and its significance on engineering design.
5. Develop understanding of dynamic balancing, flywheel analysis, gyroscopic forces and moments.

Course Outcomes:

Upon successful completion of this course the student should be able to:

1. Analyze stabilization of sea vehicles, aircrafts and automobile vehicles.
2. Compute frictional losses, torque transmission of mechanical systems.
3. Understand how to determine the natural frequencies of continuous systems starting from the general equation of displacement.
4. Understand the importance of governors, bearings, clutches and their applications
5. Understand balancing of reciprocating and rotary masses

SYLLABUS**UNIT-I**

Gyroscopic Couple and Precessional Motion: Precessional and angular motion- gyroscopic couple- effect of gyroscopic couple on an aero plane and on a naval ship, stability of a four wheel vehicle moving in a curved path, stability of a two-wheel vehicle taking a turn.

UNIT-II

Balancing of Rotating and Reciprocating Masses: Balancing of a single rotating mass in the same plane and by two masses in different planes, balancing of several masses revolving in the same plane- Balancing of several masses revolving in different planes- Primary and secondary unbalanced forces of reciprocating masses, Partial balancing of unbalanced primary forces in a reciprocating engine, Partial balancing of locomotives- Effect of partial balancing of reciprocating parts of two cylinder locomotives- Variation of tractive force, Swaying couple and hammer blow- Balancing of primary and secondary forces in multi cylinder in-line engines- Direct and reverse cranks- Balancing of V- Engines.

UNIT-III

Vibrations: Definitions- Types of vibrations- Natural frequencies of free longitudinal vibrations of systems having single degree of freedom- Equilibrium method- Energy method and Rayleigh's method. Frequency of damped vibration and forced vibration with damping- , Simple problems on forced damped vibration, Magnification factor . Natural frequency of free transverse vibrations due to point load and uniformly distributed load acting over a simply supported shaft- Transverse vibrations for a shaft subjected to number of point loads- Energy method- Dunkerley's method, Critical speed of a shaft. Natural frequency of free torsional vibrations- Free torsional vibrations of single rotor system, two rotor system and torsionally equivalent system

UNIT-IV

FRICTION: Introduction; Kinds of friction; Laws of friction; Coefficient of friction;; Screw jack; friction of screw ,Pivots and collars;; Rolling friction; Antifriction bearings; Greasy friction; Greasy friction at a journal;; Film friction;

CLUTCHES: Friction clutches- single disc or plate clutch, multiple disc clutch, cone clutch, centrifugal clutch.

UNIT-V

TURNING MOMENT DIAGRAMS: Analytical method for velocity and acceleration of piston, angular velocity and acceleration of connecting rod, engine force analysis crank effort and turning moment diagrams – fluctuation of energy – fly wheels.

GOVERNERS: Watt, porter and proell governors, spring loaded governors – Hartnell and Hartung with auxiliary springs. sensitiveness, isochronism and hunting Stability;

Text Books:

1. Theory of Machines by S.S Ratan, Mc. Graw Hill Publications
2. Theory of machines by Khurmi and S.Chand.
3. Mechanism and machine theory by Ashok G. Ambedkar, PHI Publications

Reference Books:

1. Mechanism and Machine Theory by JS Rao and RV Duggipati, New Age.
2. Theory of Machines by Shigley, MGH
3. Theory of Machines by Thomas Bevan, CBS Publishers

MACHINE DESIGN

Lecture : 3 Periods
Tutorial : 1 Period.
Exam : 3 Hrs.

Int. Marks : 30
Ext. Marks : 70
Credits : 3

Course Objectives:

1. To acquaint with standards like ASTM, ASME etc.,
2. To Design and formulate, to analyze stresses and strains in machine elements like gears, clutches, bearings, etc. under static and dynamic load conditions.
3. To Design of Internal combustion engine parts like cylinder, pistons and connecting rod
4. To understand the stresses on miscellaneous machine elements like wire ropes, chain drives etc.

Course Outcomes:

Upon the completion of course student will be able to

1. Classify different types of gears and apply the design concepts to evaluate the strength of gears.
2. Design various parts of IC Engines such as cylinders, pistons and connecting rods
3. Apply the design concepts to determine the various parameters of clutches.
4. Apply the design concepts to determine the torque and dimensions related to brakes
5. Design the sliding and roller contact bearings under various environmental and service Conditions
6. Classify and Analyze different types of stresses induced in wire ropes and chain drives

SYLLABUS**UNIT-I**

Gears: Classification of gears, Standard tooth systems. Spur, Helical and Bevel gears. Terminology of each. Tooth failure. Face width and beam strength. Lewis equation. Design for dynamic and wear loads. Force analysis of Bevel gears.

UNIT-II

I.C Engine parts: Design of cylinders and heads. Design of pistons, Design of cross-head and design of connecting rods

UNIT-III

Clutches: Torque capacity of single and multi-plate clutches. Design considerations. Energy considerations and Temperature rise friction materials. Cone clutch design, Centrifugal clutches design.

Brakes: Energy equations. self locking and self energizing conditions of a brake, Band and block brakes. Internal expanding shoe brake design.

UNIT-IV

Sliding contact bearings: Lubrication modes. Temperature effect on viscosity. Journal bearing design. Bearing modulus. McKee equations. Heating of bearings. Collar and thrust bearings.

Roller and ball bearings: Static and dynamic load capacity. Equivalent bearing load. Load-life relationships. Load factor. Selection of bearings from manufacturer's catalogue.

UNIT-V

Wire ropes: construction, Classification and designation, Stresses in wire ropes, Design for service, advantage of wire ropes over fiber ropes

Chain drives: terminology and classification, Design procedure for service. Advantages and disadvantages of chain drive over rope drives.

Text Books:

1. Design of Machine Elements by V.B. Bhandari, TMH publishing Co. Ltd., New Delhi.
2. A Textbook of Machine Design by R.S.Khurmi and J.K.Gupta, Eurasia publishing house PVT Ltd, New Delhi

Reference Books:

1. Machine Design by R.K. Jain, Khanna publications.
2. Mechanical Engineering Design by Joseph E. Shingley.

ENTREPRENEURSHIP
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To develop and strengthen entrepreneurial quality and motivation in students.
2. To impart basic entrepreneurial skills and understandings to run a business efficiently and effectively.

Course Outcomes:

1. Students will be able to understand the characteristics of entrepreneur and its role in economic development.
2. Student will be able to gain comprehensive knowledge on women entrepreneurship, rural entrepreneurship and their contribution towards economic development.
3. Students will be familiarizing with project formulation and design.
4. Students will be able to familiarize with the problems and prospectus of India.
5. Student will be able to include and implement Government of India initiatives in supporting skill development programmes.

SYLLABUS

UNIT-I

Entrepreneurship: Importance and growth - Characteristics and Qualities of Entrepreneur- Role of Entrepreneurship, Ethics and Social Responsibilities. Women Entrepreneurship: Role & Importance, Problems of Women Entrepreneurs – corporate entrepreneurship – mobility of entrepreneur – entrepreneurial motivation.

UNIT-II

Training: Designing Appropriate Training Programme to inculcate Entrepreneurial Spirit - Training for New and Existing Entrepreneurs, Feedback and Performance of Trainees. Creativity and Entrepreneurship: Sources and Methods of Ideas Planning and Development of Programmes

UNIT-III

Planning and Evaluation of Projects: Growth of Firm – Project identification and selection - Factors inducing growth- - Project Feasibility Study - Post Planning of Project-Project Planning and Control.

UNIT-IV

Small and Micro Enterprises: Importance, definitions – policies and their support to MSMEs - growth and growth strategies – sickness in small business and remedies – small entrepreneurs in International business.

UNIT-V

Institutional Support to Entrepreneur and MSMEs: Role of Government - Role of IDBI, NIESBUD, SISI, DIC - Financial Institutions-Commercial Banks, Entrepreneurial Development Institutes, Universities and other Educational Institutions offering Entrepreneurial Development Programme.

Text Books:

1. Arya Kumar: “Entrepreneurship”, Pearson, Publishing House, New Delhi, 2012.
2. VSP Rao, Kuratko: “Entrepreneurship”, Cengage Learning, New Delhi.
3. K.Ramachandran: “Entrepreneurship Development”, TMH, New Delhi, 2012.

Reference Books:

1. B.Janakiram, M Rizwana: “Entrepreneurship Development” Excel Books, New Delhi, 2011.
2. Rajeev Roy: “Entrepreneurship”, Oxford University Press, New Delhi, 2012.
3. P.C.Shejwalkar: “Entrepreneurship Development”, Everest Publishing House, New Delhi, 2011.

DATABASE MANAGEMENT SYSTEM
(Common to CE & ME)
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To understand the basic concepts and the applications of database systems.
2. To master the basics of SQL and construct queries using SQL.
3. To understand the relational database design principles.
4. To become familiar with the basic issues of transaction processing and concurrency control.
5. To become familiar with database storage structures and access techniques.

Course Outcomes:

1. Demonstrate the basic elements of a relational database management system.
2. Ability to identify the data models for relevant problems.
3. Ability to design entity relationship model and convert entity relationship diagrams into RDBMS and formulate SQL queries on the data.
4. Apply normalization for the development of application software

SYLLABUS**UNIT-I**

Introduction: Database System Applications, Purpose of Database Systems, View of Data, Database Languages – DDL, DML, Relational Databases, Database Design, Data Storage and Querying, Transaction Management, Database Architecture, Data Mining and Information Retrieval, Specialty Databases, Database Users and Administrators, History of Database Systems. Introduction to Data base design: Database Design and ER diagrams, Entities, Attributes and Entity sets, Relationships and Relationship sets, Additional features of ER Model, Conceptual Design with the ER Model, Conceptual Design for Large enterprises. Relational Model: Introduction to the Relational Model, Integrity Constraints over Relations, Enforcing Integrity constraints, Querying relational data, Logical data base Design: ER to Relational, Introduction to Views, Destroying /Altering Tables and Views.

UNIT-II

Relational Algebra and Calculus: Preliminaries, Relational Algebra, Relational calculus – Tuple relational Calculus, Domain relational calculus, Expressive Power of Algebra and calculus. SQL: Queries, Constraints, Triggers: Form of Basic SQL Query, UNION, INTERSECT, and EXCEPT, Nested Queries, Aggregate Operators, NULL values Complex Integrity Constraints in SQL, Triggers and Active Data bases, Designing Active Databases..

UNIT-III

Schema Refinement and Normal Forms: Introduction to Schema Refinement, Functional Dependencies-Reasoning about FDs, Normal Forms, Properties of Decompositions, Normalization, Schema Refinement in Database Design, Other Kinds of Dependencies.

UNIT-IV

Transaction Management: Transactions, Transaction Concept, A Simple Transaction Model, Storage Structure, Transaction Atomicity and Durability, Transaction Isolation, Serializability, Transaction Isolation and Atomicity Transaction Isolation Levels, Implementation of Isolation Levels. Concurrency Control: Lock-Based Protocols, Multiple Granularity, Timestamp-Based Protocols, Validation-Based Protocols, Multiversion Schemes. Recovery System-Failure Classification, Storage, Recovery and Atomicity, Recovery Algorithm, Buffer Management, Failure with loss of nonvolatile storage, Early Lock Release and Logical Undo Operations, Remote Backup systems.

UNIT-V

Storage and Indexing: Overview of Storage and Indexing: Data on External Storage, File Organization and Indexing, Index Data Structures, Comparison of File Organizations. Tree-Structured Indexing: Intuition for tree Indexes, Indexed Sequential Access Method (ISAM), B+ Trees: A Dynamic Index Structure, Search, Insert, Delete. Hash- Based Indexing: Static Hashing, Extendible hashing, Linear Hashing, Extendible vs. Linear Hashing.

Text Books:

1. Data base Management Systems, Raghu Ramakrishnan, Johannes Gehrke, McGraw Hill Education (India) Private Limited, 3rd Edition.
2. Data base System Concepts, A. Silberschatz, Henry. F. Korth, S. Sudarshan, McGraw Hill Education(India) Private Limited l, 6th edition.

Reference Books:

1. Database Systems, 6th edition, R Elmasri, ShamkantB.Navathe, Pearson Education.
2. Database System Concepts, Peter Rob & Carlos Coronel, Cengage Learning.
3. Introduction to Database Management, M. L. Gillenson and others, Wiley Student Edition.
4. Database Development and Management, Lee Chao, Auerbach publications, Taylor & Francis Group. Introduction to Database Systems, C. J. Date, Pearson Education.

WASTE WATER MANAGEMENT
(Common to CE & ME)
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

Student shall be able to:

1. Explain the difference between domestic waste and industrial waste water
2. Understand the common methods of treatment used in industries
3. To understand the effluent quality and its impact on environment.

Course Outcomes:

By the end of the course, the student should be able to:

1. Define the quality of industrial wastes
2. Explain various industrial waste treatment processes
3. Outline the advanced treatment techniques available for industrial wastes
4. Explain the sludge reduction and disposal methodologies
5. Analyze the waste effluent treatment from different case studies

SYLLABUS**UNIT-I**

Water uses by industry – Sources of water for industries – Characteristics of industrial wastes – Industrial water requirements – quality and quantity

UNIT-II

Waste reduction – Volume reduction – Classification of wastes – Equalisation- Neutralisation – Floatation – Precipitation – Heavy metal removal - adsorption – Aerobic and anaerobic biological treatment – reed bed technology

UNIT-III

Measurement of industrial waste water flow – waste water characterization - Advanced waste reduction technologies – Ozonation – Membrane technologies – Ion exchange – Nutrient removal – recycling, reuse and resources recovery

UNIT-IV

Waste disposal methods- land treatment – water bodies, rivers, oceans – problems of disposal – Common effluent treatment plants- advantages and disadvantages – recirculation of industrial wastes – Effluent disposal methods – sludge treatment - disposal

UNIT-V

Manufacturing process and origin, characteristics, effects and treatment methods of liquid waste from different industries – steel – fertilizers – textiles – paper and pulp industries – oil refineries – coal and gas power plants- tanneries – sugar – textiles – distillery – dairy – food processing - distilleries

Text Book:

1. Eckenfelder, W.W., (2000)'Industrial Water Pollution Control", Mc-Graw Hill.

Reference Book:

1. Industrial waste water treatment by A.D. Patwardhan

**COMPUTER GRAPHICS
(Open Elective)**

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. Provides a comprehensive introduction to computer graphics with a foundation in Graphics Applications.
2. A thorough introduction to computer graphics techniques.
3. To give the basics of Geometric Transformations and projections.

Course Outcomes:

1. The students will understand graphics principles and graphics hardware.
2. The students can demonstrate geometrical transformations.
3. The students can create interactive graphics applications and demonstrate computer graphics

SYLLABUS

UNIT I

Introduction: Computer Graphics and their applications: Computer Aided Design, Computer Art, Entertainment, Education and Training, Graphical User Interfaces; Overview of Graphics systems: Video Display Devices, Raster Scan Systems, Random Scan Systems, Graphics Monitors And Workstations, Input Devices, Hard Copy Devices, Interactive Input Methods, Windows and Icons, Virtual Reality Environments, Graphics Software.

UNIT II

Output primitives :Points and Lines, , Line and Curve Attributes, Color and Gray scale levels, Antialiasing, Loading the Frame buffer, Line function, Line Drawing Algorithms, Circle Generating Algorithms, Ellipse Generating Algorithms, Pixel Addressing, Area Fill Attributes, Filled Area Primitives, Filled Area Functions, Cell Array, Character Generation, Character Attributes, Bundled Attributes, Curve Functions, Parallel Curve Algorithms.

UNIT-III

Two Dimensional Transformations: Basic 2D Transformations, Matrix Representations, Homogeneous Coordinates, Composite Transformations, Other Transformations, Transformations between Coordinate Systems, Affine Transformations.

Viewing Pipeline and Clipping operations : Viewing Pipeline ,Viewing Coordinates & Reference frames, Window-to-Viewport Coordinate Transformation, Two Dimensional Viewing Functions, Clipping and its Operations, Types of clipping operations- Point Clipping, Line Clipping, Polygon Clipping,, Curve Clipping,, Text and Exterior Clipping.

UNIT-IV

Three Dimensional Transformations & Projections: Translation, Rotation, Scaling, Other Transformations, Composite Transformations, 3D Transformation Functions, Modeling and Coordinate Transformations, Need for projections, Parallel & Perspective projections, General Projection Transformations.

UNIT-V

Three Dimensional Concepts and Object representations: 3D display methods, 3D Graphics, Polygon Surfaces, Curved Lines and Surfaces, Quadratic Surfaces, Super Quadrics, Blobby Objects, Spline Representations, Bézier Curves and Surfaces, BSpline Curves and Surfaces.

Text Books:

1. Computer Graphics, Donald Hearn & M.Pauline Baker, Pearson Education ,New Delhi.
2. Procedural Elements for Computer Graphics, David F.Rogers, Tata Mc Graw Hill Book Company, NewDelhi, 2003.

Reference Books:

1. Computer Graphics: Principles & Practice in C, J.D. Foley, S.K Feiner, A Van Dam F.H John Pearson Education, 2004.
2. Computer Graphics using Open G L Francis Shill Jr, Pearson Education, 2004.
3. Computer Vision and Image Processing: A Practical Approach using CVIP tools, S. E. Umbaugh, Prentice Hall, 1998.

INDUSTRIAL ROBOTICS
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

1. To give students practice in applying their knowledge of mathematics, science, and Engineering and to expand this knowledge into the vast area of robotics.
2. The students will be exposed to the concepts of robot kinematics and dynamics
3. Mathematical approach to explain how the robotic arm motion can be described.
4. The students will understand the functioning of sensors and actuators.

Course Outcomes:

Students will be able to

1. Distinguish between fixed automation and programmable automation.
2. Identify various components of robot.
3. Select appropriate type of actuator for a joint.
4. Illustrate robot applications in manufacturing.
5. Analyze kinematics of a robot.

SYLLABUS

UNIT - I

INTRODUCTION: Automation and Robotics, Robot anatomy, robot configuration, motions joint notation scheme, work volume, robot drive systems, control systems and dynamic performance, precision of movement.

CONTROL SYSTEM AND COMPONENTS: basic concepts and motion controllers, control system analysis, robot actuation and feedback components, Positions sensors, velocity sensors, actuators, power transmission systems, robot joint control design.

UNIT - II

MOTION ANALYSIS AND CONTROL: Manipulator kinematics, position representation, forward and inverse transformations, homogeneous transformations, manipulator path control, robot arm dynamics, configuration of a robot controller.

UNIT - III

END EFFECTORS: Grippers-types, operation, mechanism, force analysis, tools as end effectors consideration in gripper selection and design. **SENSORS:** Desirable features, tactile, proximity and range sensors, uses sensors in robotics.

MACHINE VISION: Functions, Sensing and Digitizing-imaging devices, Lighting techniques, Analog to digital single conversion, image storage: Image processing and Analysis-image data reduction, Segmentation, feature extraction, Object recognition. Training the vision system, Robotic application.

UNIT - IV

ROBOT PROGRAMMING: Lead through programming, Robot program as a path in space, Motion interpolation, WAIT, SIGNAL AND DELAY commands, Branching, capabilities and Limitations of lead through methods.

ROBOT LANGUAGES: Textual robot Languages, Generations of robot programming languages, Robot language structures, Elements and function.

UNIT - V

ROBOT CELL DESIGN AND CONTROL: Robot cell layouts-Robot centered cell, In-line robot cell, Considerations in work design, Work and control, Inter locks, Error detection, Work cell controller.

ROBOT APPLICATION: Material transfer, Machine loading/unloading, Processing operation, Assembly and Inspection, Future Application.

Text Books:

1. Industrial Robotics /Groover M P /Pearson Edu.
2. Introduction to Robotic Mechanics and Control by JJ Craig, Pearson, 3rd edition.

Reference Books:

1. Robotics / Fu K S/ McGraw Hill.
2. Robotic Engineering / Richard D. Klafter, Prentice Hall.
3. Robot Analysis and Intelligence / Asada and Slotine / Wiley Inter-Science.
4. Robot Dynamics & Control – Mark W. Spong and M. Vidyasagar / John Wiley
5. Introduction to Robotics by SK Saha, The McGraw Hill Company, 6th, 2012.
6. Robotics and Control / Mittal R K & Nagrath I J / TMH.

GREEN ENGINEERING SYSTEMS
(Open Elective)

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The course aims to highlight the:

1. Significance of alternative sources of energy
2. Significance of green energy systems and processes and provides the theory and working principles of probable sources of renewable and green energy systems that are environmental friendly.

Course Outcomes:

The student shall:

1. Understand the principles and working of solar energy and solar energy solar energy collection.
2. Understand the principles of solar energy storage and applications of solar energy and wind energy.
3. Understand the principles and working of biomass, geo thermal and ocean energies and appreciate their significance in view of their importance in the current scenario and their potential future applications.
4. Understand the principles and working of energy efficient systems like electrical and mechanical systems.
5. Understand the principles and working of energy efficient processes.

SYLLABUS

UNIT-I

INTRODUCTION: SOLAR RADIATION: Role and potential of new and renewable sources, the solar energy option, Environmental impact of solar power, structure of the sun, the solar constant, sun-earth relationships, coordinate systems and coordinates of the sun, extraterrestrial and terrestrial solar radiation, solar radiation on tilted surface, instruments for measuring solar radiation and sun shine, solar radiation data, numerical problems. Photo voltaic energy conversion – types of PV cells, I-V characteristics. **SOLAR ENERGY COLLECTION:** Flat plate and concentrating collectors, classification of concentrating collectors, orientation and thermal analysis, advanced collectors.

UNIT-II

SOLAR ENERGY STORAGE AND APPLICATIONS: Different methods, sensible, latent heat and stratified storage, solar ponds, solar applications- solar heating/cooling technique, solar distillation and drying, solar cookers, central power tower concept and solar chimney. **WIND ENERGY:** Sources and potentials, horizontal and vertical axis windmills, performance characteristics, betz criteria, types of winds, wind data measurement.

UNIT-III

BIO-MASS: Principles of bio-conversion, anaerobic/aerobic digestion, types of bio-gas digesters, gas yield, combustion characteristics of bio-gas, utilization for cooking, bio fuels, I.C. engine operation and economic aspects. **GEOTHERMAL ENERGY:** Resources, types of wells, methods of harnessing the energy, potential in India. **OCEAN ENERGY:** OTEC, Principles of utilization, setting of OTEC plants, thermodynamic cycles. Tidal and wave energy: Potential and conversion techniques, mini-hydel power plants, and their economics.

UNIT-IV

ENERGY EFFICIENT SYSTEMS: ELECTRICAL SYSTEMS: Energy efficient motors, energy efficient lighting and control, selection of luminaire, variable voltage variable frequency drives (adjustable speed drives), controls for HVAC (heating, ventilation and air conditioning), demand site management. **MECHANICAL SYSTEMS:** Fuel cells- principle, thermodynamic aspects, selection of fuels & working of various types of fuel cells, Environmental friendly and Energy efficient compressors and pumps.

UNIT-V

ENERGY EFFICIENT PROCESSES: Environmental impact of the current manufacturing practices and systems, benefits of green manufacturing systems, selection of recyclable and environment friendly materials in manufacturing, design and implementation of efficient and sustainable green production systems with examples like environmental friendly machining, vegetable based cutting fluids, alternate casting and joining techniques, zero waste manufacturing.

Text Books:

1. Sukhatme S.P. and J.K.Nayak, Solar Energy – Principles of Thermal Collection and Storage, TMH.
2. Khan B.H., Non-Conventional Energy Resources, Tata McGraw Hill, New Delhi, 2006.
3. Green Manufacturing Processes and Systems, Edited by J. Paulo Davim, Springer 2013.

Reference Books:

1. Alternative Building Materials and Technologies / K.S Jagadeesh, B.V Venkata Rama Reddy and K.S Nanjunda Ra.
2. Principles of Solar Energy / Frank Krieth& John F Kreider.
3. Non-Conventional Energy / Ashok V Desai /Wiley Eastern.
4. Renewable Energy Technologies /Ramesh & Kumar /Narosa
5. Renewable Energy Technologies/ G.D Roy

COMPUTER AIDED DESIGN

Lecture	: 3 Periods	Int. Marks	: 30
Tutorial	: 1 Period.	Ext. Marks	: 70
Exam	: 3 Hrs.	Credits	: 3

Course Objectives:

The overall objective of the course is to teach the basic principles of CAD which includes computer graphics, application of computers in FEM and development algorithms of different mechanical elements. Specific objectives include:

1. This course integrates theoretical approaches and advanced tools for geometric modelling in mechanical engineering applications.
2. This course will help students actively involve to build on practical techniques in exploring issues of current interest in engineering computer aided design and analysis.
3. Student will be using advanced modelling approaches and tools for real - world mechanical design problems that will enable him to execute comprehensive and professional engineering projects.
4. Through topics in this course addressing the impact of CAD in engineering design and analysis, student will develop the ability to establish the CAD techniques appropriate for mechanical engineering applications.
5. The interdisciplinary nature of computer graphics, geometric modelling and engineering design is emphasized in the wide variety of examples and applications presented with both broad work and addressed through the hands - on nature of the course work.
6. The course material is structured to meet the needs of Automotive, Mechanical and manufacturing engineering students focussing on CAD and its applications to problem solving in engineering design and analysis.

Course Outcomes:

Upon completion of this course, students should be able:

1. Analyze and use engineering computer graphics and geometric modelling techniques for mechanical engineering applications.
2. Able to understand and apply theories, methods and procedures for complex-shapes part design.
3. Apply advanced modelling and computational tools for complex part and shape design and analysis.
4. Select and use various engineering design procedures for mechanical design problems involving complex shapes.
5. Execute professional engineering CAD projects for mechanical engineering applications in the current industrial practice.

SYLLABUS

UNIT-I

Fundamentals of CAD - Introduction - The design process - Application of computers for design - Operating systems - Hardware in CAD: The design work station - I/O Devices - CAD system configuration - Creating database for manufacturing - Benefits of CAD.

UNIT-II

Interactive Computer Graphics - Graphic display devices- Graphics system- Graphics standards - Graphical user interface- Transformation systems- windowing - clipping - 2D and 3D transformations - Linear transformation- Geometric Modeling - Modeling Techniques - Wire frame Modeling - Surface Modeling - 3 D Solid Modeling.

UNIT-III

Introduction to Finite Element Analysis - CAD techniques to finite element data preparation- Automatic mesh generation- presentation of results - 3-dimensional shape description and mesh generation- CAD applications of FEM.

UNIT-IV

CAD applications and exposure to CAD packages: Simple examples of computer aided drafting, design and analysis - Introduction to simple machine elements - Analysis of cross sectional area, centroid & moment of inertia- Kinematics of crank- slider mechanism and other simple design applications. Introduction to CAD packages like ANSYS, NASTRAN, NISA-II.

UNIT-V

Introduction to Artificial Intelligence - Applications of AI in design and CAD.

Expert system: Structure and characteristics of expert system, building an expert system, knowledge representation, benefits of expert system.

Text Books:

1. CAD/CAM- Computer Aided Design & Manufacturing, by M.D.Groover & E.W.Zimmer.
2. CAD/CAM/CIM by Radhakrishna, New age international

Reference Books:

1. Computer Aided Design in Mechanical Engineering, by V.Rama Murthy.
2. Elements of Computer Aided Design & Manufacturing, by Y.C.Pao.
3. Computer Aided Kinetics for Machine Design, by D.L.Ryan.
4. Computer Aided Design and Manufacturing, by C.B.Besant & C.W.K.Lui.
5. Computer-Aided Analysis & Design by S. Ghosal, Prentice Hall of India.
6. CAD CAM by IBRAHIM ZEID

INDUSTRIAL ENGINEERING LAB**Lab : 3 Periods****Exam : 3 Hrs.****Int. Marks : 50****Ext. Marks : 50****Credits : 2**

Course Objectives:

1. To make student acquainted with the control charts and measure the quality of the product.
2. To make aware of the different types of process charts used for improving the method of doing the work.
3. To help students to learn the different methods of finding the standard time for a job.
4. To make the students acquainted with the probability distributions.
5. To make the students learn the impact of work on the human physiology and the physiological constraints of the body.

Course Outcomes:

1. Students will be able to find the quality of the product using different charts.
2. Can improve the method of doing work by applying principle of motion economy and method study charts.
3. Can find the standard time required for completing a job by different methods.
4. Understands the basic probability distributions.
5. Understands the impact of work on the human body and also the physiological constraints of the body

LIST OF EXPERIMENTS

1. To measure the skill and dexterity in the movement of Wrist and Fingers using pin board.
2. To measure the Heart beat using Stethoscope.
3. To show that the sample means from a normal universe follow a normal distribution.
4. To draw the control chart for fraction defective for a given lot of marble balls.
5. To determine the cycle time using PMTS.
6. To draw two handed process charts for (i). Bolt, Washer and nut assembly; (ii). Assembly of electric tester.
7. To study the changes in heart rate for different subjects using Tread mill.
8. To draw Multiple Activity chart using an electric toaster.
9. To determine the percentage utilization using work sampling.
10. To study the process capability of a given process.
11. To measure the Heart rate during working and recovery periods of the subjects under different loads, using Bicycle ergometer.
12. To draw flow process charts on activities in Workshop/ Laboratory/Office.

13. To determine the time required to perform motion sequence using work factor system.
14. To draw SIMO charts for (i). Ball point pen assembly; (ii). Electric plug assembly.
15. To conduct time study of the bulb holder assembly operation of the existing method.
16. To collect the anthropometrics data using 'Anthropolometer'.

Reference Books:

1. Industrial Engineering Management, by Dr. O. P .Khanna.
2. Industrial Engineering and Production Management by Telsay, S. Chand & Co.

FLUID MECHANICS AND MACHINERY LAB

Lab : 3 Periods

Exam : 3 Hrs.

Int. Marks : 50

Ext. Marks : 50

Credits : 2

Course Objectives:

1. To understand the working principles and fundamentals of Fluid mechanics, turbines and pumps.
2. To prepare the student for industrial applications.
3. The students are taught with testing and performance of hydraulic machines.

Course Outcomes:

1. The student gets complete knowledge on fluid mechanics, hydraulic turbines and pumps.
2. The student learns the complete calculation procedures for designing hydraulic turbines, and pumps.
3. The student is prepared to work in industry immediately after this course.

LIST OF EXPERIMENTS

1. Calibration of flow meters: (a) Venturi meter; (b) Orifice meter; (c) Nozzle meter.
2. Determination of coefficient of discharge for: (a) small orifice; (b) cylindrical mouth piece.
3. Finding coefficient of discharge for: (a) rectangular notch; (b) triangular notch; (c) trapezoidal notch.
4. To draw the performance characteristics of Centrifugal pump.
5. To find the specific speed of: (a) Pelton turbine; (b) Francis turbine.
6. To draw the characteristic curves for reciprocating pump.
7. To draw the pressure distribution and finding coefficient of drag for a bluff body and an Aero foil.
8. To draw the characteristic curves for the hydraulic ram.

Reference Books:

1. Fluid Mechanics and Hydraulic Machinery, by R.K.Bansal, Laxmi publications.
2. Fluid Mechanics and Hydraulic Machinery, by R.K.RAJPUT.
3. Fluid Flow Machines, by N.S.Govinda Rao, Tata McGraw Hill publishing company Ltd.
4. "Hydraulics & Fluid Mechanics", P.N. Modi and S.M. Seth, TEXT BOOKS House, Delhi.

EMPLOYABILITY SKILLS
(Common to all Branches)

Tutorial : 3 Periods (VA-2+QA-1)
Exam : 3 Hrs.

Int.Marks : 30
Ext.Marks : 70
Credits : 1

Part-A: Verbal Aptitude and Soft Skills-II

Course objectives:

1. To expose the students to bettering sentence expressions and also forming equivalents.
2. To instill reading and analyzing techniques for better comprehension of written discourses.
3. To create awareness among the students on the various aspects of writing, organizing data, preparing reports, and applying their writing skills in their professional career.
4. To inculcate conversational skills, nuances required when interacting in different situations.
5. To build/refine the professional qualities/skills necessary for a productive career and to instill confidence through attitude building.

Course Outcomes:

The students will be able to

1. Construct coherent, cohesive and unambiguous verbal expressions in both oral and written discourses.
2. Analyze the given data/text and find out the correct responses to the questions asked based on the reading exercises; identify relationships or patterns within groups of words or sentences
3. Write paragraphs on a particular topic, essays (issues and arguments), e mails, summaries of group discussions, reports, make notes, statement of purpose(for admission into foreign universities), letters of recommendation(for professional and educational purposes).
4. Converse with ease during interactive sessions/seminars in their classrooms, compete in literary activities like elocution, debates etc., raise doubts in class, participate in JAM sessions/versant tests with confidence and convey oral information in a professional manner.
5. Participate in group discussions/group activities, exhibit team spirit, use language effectively according to the situation, respond to their interviewer/employer with a positive mind, tailor make answers to the questions asked during their technical/personal interviews, exhibit skills required for the different kinds of interviews (stress, technical, HR) that they would face during the course of their recruitment process.

SYLLABUS

UNIT -I (VA)

Sentence Improvement (finding a substitute given under the sentence as alternatives), Sentence equivalence (completing a sentence by choosing two words either of which will fit in the blank), cloze test (reading the written discourse carefully and choosing the correct options from the alternatives and filling in the blanks), summarizing and paraphrasing.

UNIT- II (VA)

Types of passages (to understand the nature of the passage), types of questions (with emphasis on inferential and analytical questions), style and tone (to comprehend the author's intention of

writing a passage), strategies for quick reading(importance given to skimming, scanning), summarizing ,reading between the lines, reading beyond the lines, techniques for answering questions related to vocabulary (with emphasis on the context), supplying suitable titles to the passage, identifying the theme and central idea of the given passages.

UNIT- III (VA)

Punctuation, discourse markers, general Essay writing, writing Issues and Arguments(with emphasis on creativity and analysis of a topic), paragraph writing, preparing reports, framing a ‘_Statement of purpose’, ‘_Letters of Recommendation’, business letter writing, email writing, writing letters of complaints/responses. picture perception and description, book review.

UNIT-IV (VA)

Just a minute sessions, reading news clippings in the class, extempore speech, telephone etiquette, making requests/suggestions/complaints, elocutions, debates, describing incidents and developing positive non verbal communication, story narration, product description.

UNIT-V (SS)

Employability Skills – Significance — Transition from education to workplace - Preparing a road map for employment – Getting ready for the selection process, Awareness about Industry / Companies – Importance of researching your prospective workplace - Knowing about Selection process - Resume Preparation: Common resume blunders – tips, Resume Review, Group Discussion: Essential guidelines – Personal Interview: Reasons for Rejection and Selection.

Reading/ Listening material:

1. Guide to IELTS, Cambridge University Press
2. Barron’s GRE guide.
3. Newspapers like ‘_The Hindu, _Times of India, _Economic Times.
4. Magazines like Frontline, Outlook and Business India.
5. News channels NDTV, National News, CNN

Text Books:

1. Objective English and Verbal Reasoning by R S Agarwal.
2. Communication Skills by Sanjay Kumar and Pushpa Latha, Second Edition, OUP.
3. Business Correspondence and Report Writing – A Practical Approach to Business and Technical Communication by R C Sharma and Krishna Mohan.
4. Soft Skills & Employability Skills by Samina Pillai and Agna Fernandez, Cambridge University Press India Pvt. Ltd.
5. Soft Skills, by Dr. K. Alex, S. Chand & Company Ltd., New Delhi

Reference Books:

1. Oxford Guide to Effective Writing and Speaking by John Seely.
2. Collins Cobuild English Grammar by Collins
3. The Art of Public Speaking by Dale Carnegie
4. The Leader in You by Dale Carnegie
5. Emotional Intelligence by Daniel Golman
6. Stay Hungry Stay Foolish by Rashmi Bansal

Part-B: Quantitative Aptitude-II

Course objectives:

The objective of introducing quantitative aptitude-II is:

1. To refine concepts related to quantitative aptitude. – SOLVING PROBLEMS OF DI and accurate values using averages, percentages.
2. To inculcate logical thinking by exposing the students to puzzles and reasoning related questions.
3. To familiarize the students with finding out accurate date and time related problems.
4. To enable the students solve the puzzles using logical thinking.
5. To expose the students to various problems based on geometry and mensuration.

Course Outcomes:

1. The students will be able to perform well in calculating different types of data interpretation problems.
2. The students will perform efficaciously on analytical and logical problems using various methods.
3. Students will find the angle measurements of clock problems with the knowledge of calendars and clock.
4. The students will skillfully solve the puzzle problems like arrangement of different positions.
5. The students will become good at solving the problems of lines, triangulars, volume of cone, cylinder and so on.

SYLLABUS

UNIT I: Averages, mixtures and allegations, Data interpretation Understanding of AM,GM,HM-Problems on averages, Problems on mixtures standard method. Importance of data interpretation: Problems of data interpretation using line graphs, Problems of data interpretation using bar graphs, Problems of data interpretation using pie charts, Problems of data interpretation using others.

UNIT II: Puzzle test, blood Relations, permutations, Combinations and probability Importance of puzzle test, Various Blood relations-Notation to relations and sex making of family Tree diagram, Problems related to blood relations, Concept of permutation and combination, Problems on permutation, Problems on combinations, Problems involving both permutations and combinations, Concept of probability-Problems on coins, Problems on dice, Problems on cards, Problems on years.

UNIT III: Periods,Clocks, Calendars, Cubes and cuboids Deriving the formula to find the angle between hands for the given time, finding the time if the angle is known, Faulty clocks, History of calendar-Define year, leap year, Finding the day for the given date, Formula and method to find the day for the given date in easy way, Cuts to cubes, Colors to cubes, Cuts to cuboids, Colors to cuboids.

UNIT IV: Puzzles Selective puzzles from previous year placement papers, sitting arrangement, problems- circular arrangement, linear arrangement, different puzzles.

UNIT V: Geometry and Mensuration Introduction and use of geometry-Lines, Line segments, Types of angles, Intersecting lines, Parallel lines, Complementary angles, supplementary angles, Types of triangles-Problems on triangles, Types of quadrilaterals-Problems on quadrilaterals,

Congruent triangles and properties, Similar triangles and its applications, Understanding about circles-Theorems on circles, Problems on circles, Tangents and circles, Importance of mensuration-Introduction of cylinder, cone, sphere, hemi sphere.

Text Books:

1. Quantitative aptitude by RS Agarwal
2. Verbal and non verbal reasoning by RS Agarwal.
3. Puzzles to puzzle you by shakunataladevi
4. More puzzles by shakunataladevi
5. Puzzles by George summers.

Reference Books:

1. Barron's by Sharon Welner Green and Ira K Wolf (Galgotia Publications pvt. Ltd.)
2. Websites: m4maths, Indiabix, 800score, official CAT, GRE and GMAT sites
3. Material from _IMS, Career Launcher and Time' institutes for competitive exams.
4. Books for cat by arunsharma
5. Elementary and Higher algebra by HS Hall and SR knight.

BASIC CODING
(Common to CE & ME)

Lab	: 3 Periods	Int.Marks	: 50
Exam	: 3 Hrs.	Ext. Marks	: 50
		Credits	: 1

Course Objectives:

1. To develop programming skills among the students.
2. To familiarize the student with Control Structures, Loop Structures.
3. To familiarize the student with Basic searching and sorting Methods.
4. To familiarize the student with Functions, Recursions and Storage Classes.
5. To familiarize the student with Structures and Unions.
6. To familiarize the student with Operating System concepts.
7. To familiarize the student with Networking concepts.

Course Outcomes:**At the end of the course students will be able to**

1. Know about Control Structures, Loop Structures and branching in programming.
2. Know about various searching and sorting methods.
3. Know about Functions, Recursions and Storage Classes.
4. Know about Structures and Unions.
5. Know different Operating System concepts.
6. Differentiate OSI Model Vs. TCP/IP suite.

SYLLABUS

UNIT I Review of Programming constructs

Programming Environment, Expressions formation, Expression evaluation, Input and Output patterns, Control Structures, Sequential branching, Unconditional branching, Loop Structures, Coding for Pattern Display.

UNIT II Introduction to Linear Data, strings and pointers

Structure of linear data, Operation logics, Matrix forms and representations, Pattern coding, Working on character data, Compiler defined methods, Substitution coding for defined methods, Row Major representation, Column Major representation, Basic searching and sorting Methods.

UNIT III Functions, Recursions and Storage Classes

Functions – Introduction to modular programming – Function Communication - Pass by value, Pass by reference – Function pointers – Recursions – Type casting – Storage classes

Practice: programs on passing an array and catching by a pointer, function returning data, comparison between recursive and Iterative solutions.

Data referencing mechanisms: Pointing to diff. data types, Referencing to Linear data, Runtime-memory allocation, Named locations Vs pointed locations, Referencing a 2D-Matrix.

UNIT IV User-defined data types, Pre-processor Directives and standard storage

Need for user-defined data type – structure definition – Structure declaration – Array within a Structure – Array of Structures – Nested Structures - Unions – Declaration of Union data type, Struct Vs Union - Enum – Pre-processor directives , Standard storage methods, Operations on file, File handling methods, Orientation to Object oriented programming

Practice: Structure padding, user-defined data storage and retrieval programs

UNIT V Operating system principles and Database concepts

Introduction to Operating system principles, Process scheduling algorithms, Deadlock detection and avoidance, Memory management, Networking: Introduction to Networking, OSI Model Vs. TCP/IP suite, Datalink layer, Internet layer, DVR Vs. LSR, Transport Layer, Application Layer

References:

1. Computer Science, A structured programming approach using C, B.A.Forouzan and R.F.Gilberg, 3rd Edition, Thomson, 2007.
2. The C –Programming Language, B.W. Kernighan, Dennis M. Ritchie, Prentice Hall India Pvt.Ltd
3. Scientific Programming: C-Language, Algorithms and Models in Science, Luciano M. Barone (Author), Enzo Marinari (Author), Giovanni Organtini, World Scientific .
4. Object Oriented Programming in C++: N. Barkakati, PHI.
5. Object Oriented Programming through C++ by Robat Laphore.
6. <https://www.geeksforgeeks.org/>.
7. <https://www.tutorialspoint.com/>